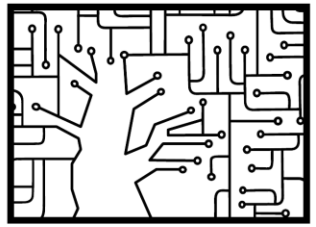
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Deep Learning for Computer Vision: **Object Detection & Segmentation**

Shai Bagon

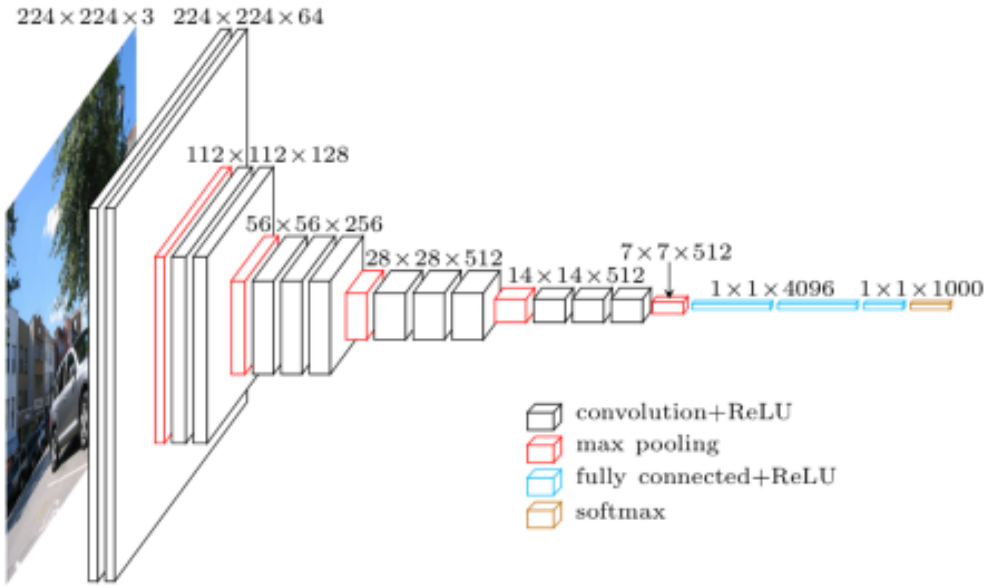


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Recap: Deep Learning

Key ingredients

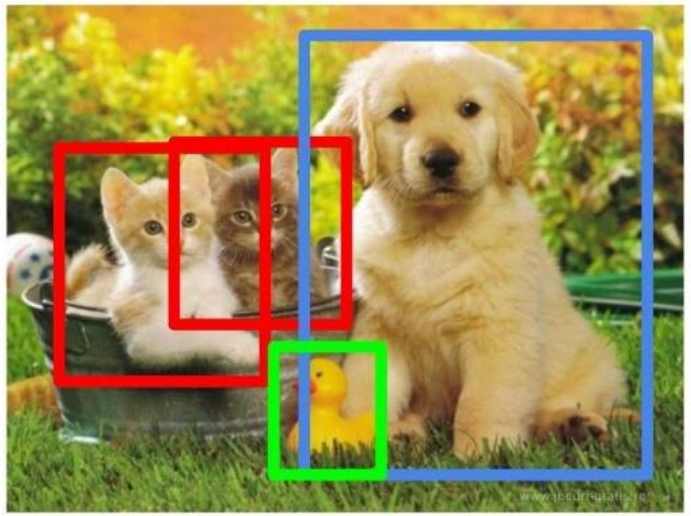
- **Data:** labeled examples $\{(\boldsymbol{x}_i, y_i)\}_{i=1\ldots N}$
- **Model:** a "deep net" $\hat{y} = f(\boldsymbol{x}; \boldsymbol{\theta})$
- **Criterion:** how to "match" model and data $\mathcal{L}(\hat{y}, y)$
- **Optimization:** Stochastic Gradient Descent (SGD)



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Additional Tasks

- Object detection
- Semantic segmentation
- Instance segmentation







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Additional Tasks: Training Data

MS COCO





- 200K labeled images
- 1.5**M** instances
- 80 object categories

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Object Detection

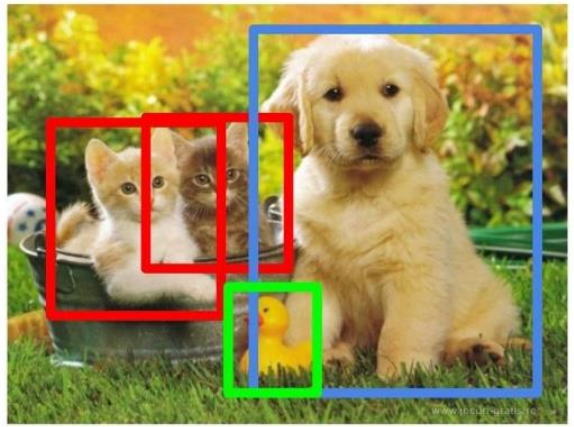
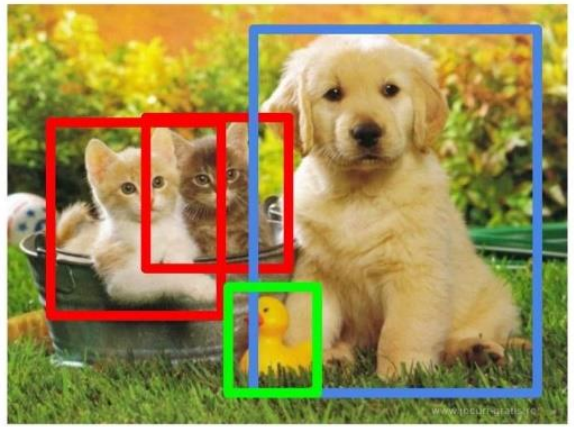


Image credit: medium

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Object Detection - Challenges

- Multiple types of outputs
- Varying number of objects



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Localization

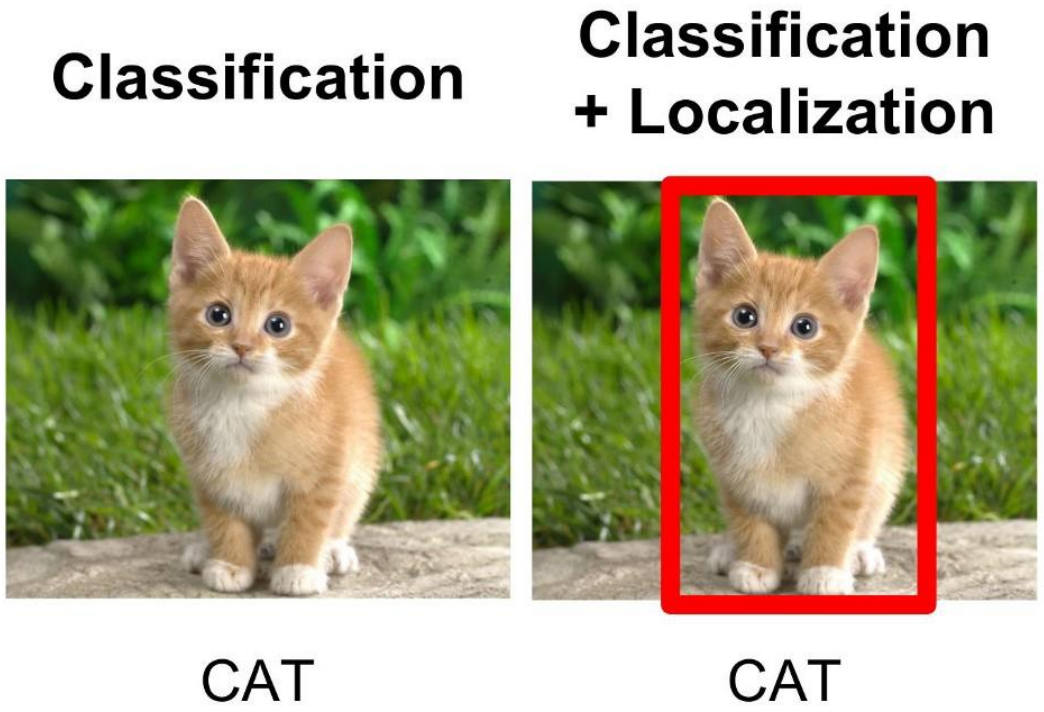
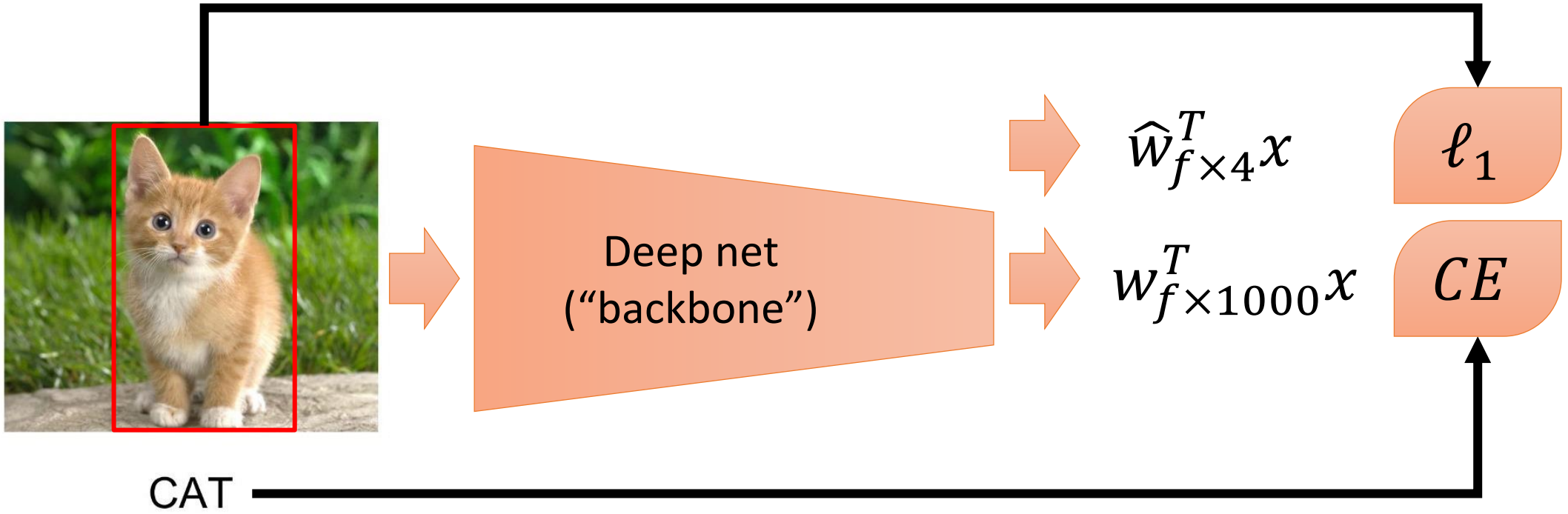


Image credit: medium

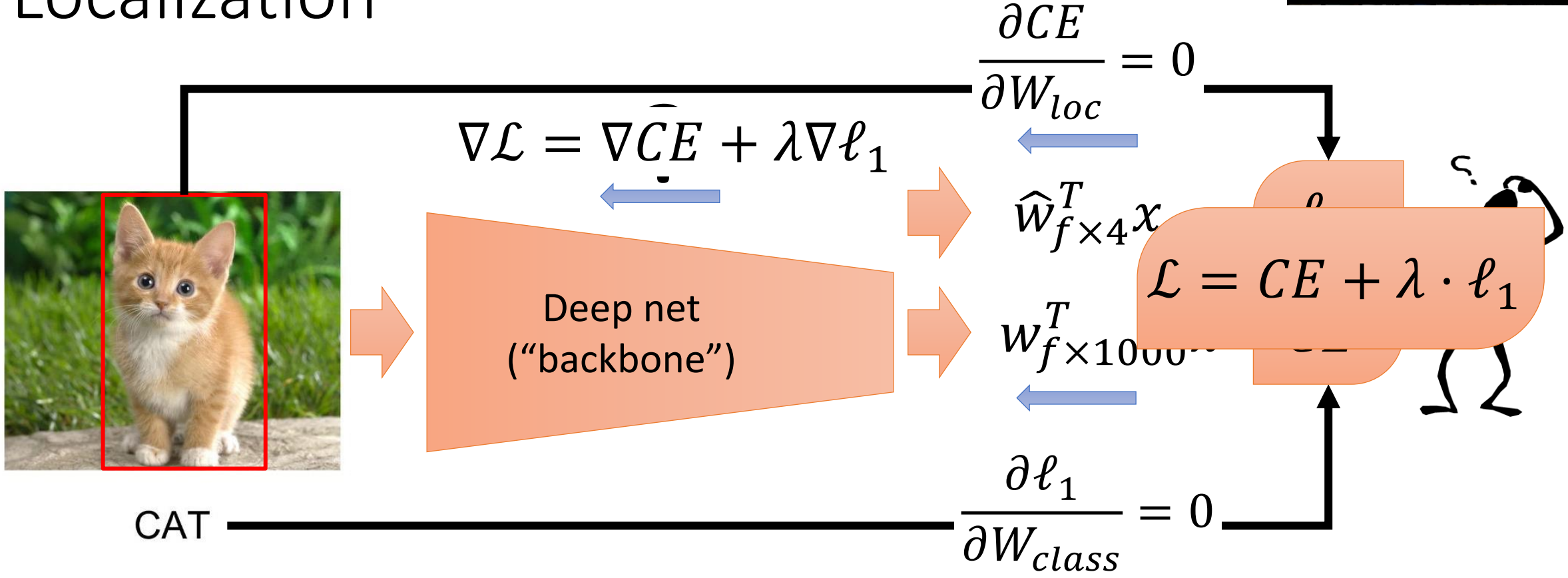
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Localization



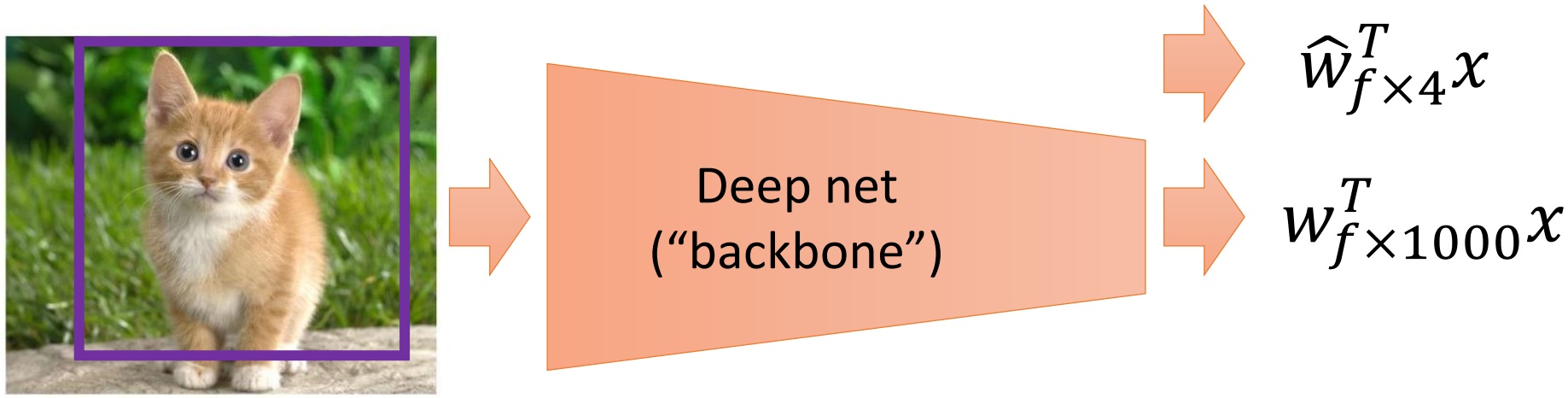
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Localization



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From Localization to Detection (v0)



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From Localization to Detection (v0)



Deep net
(“backbone”)

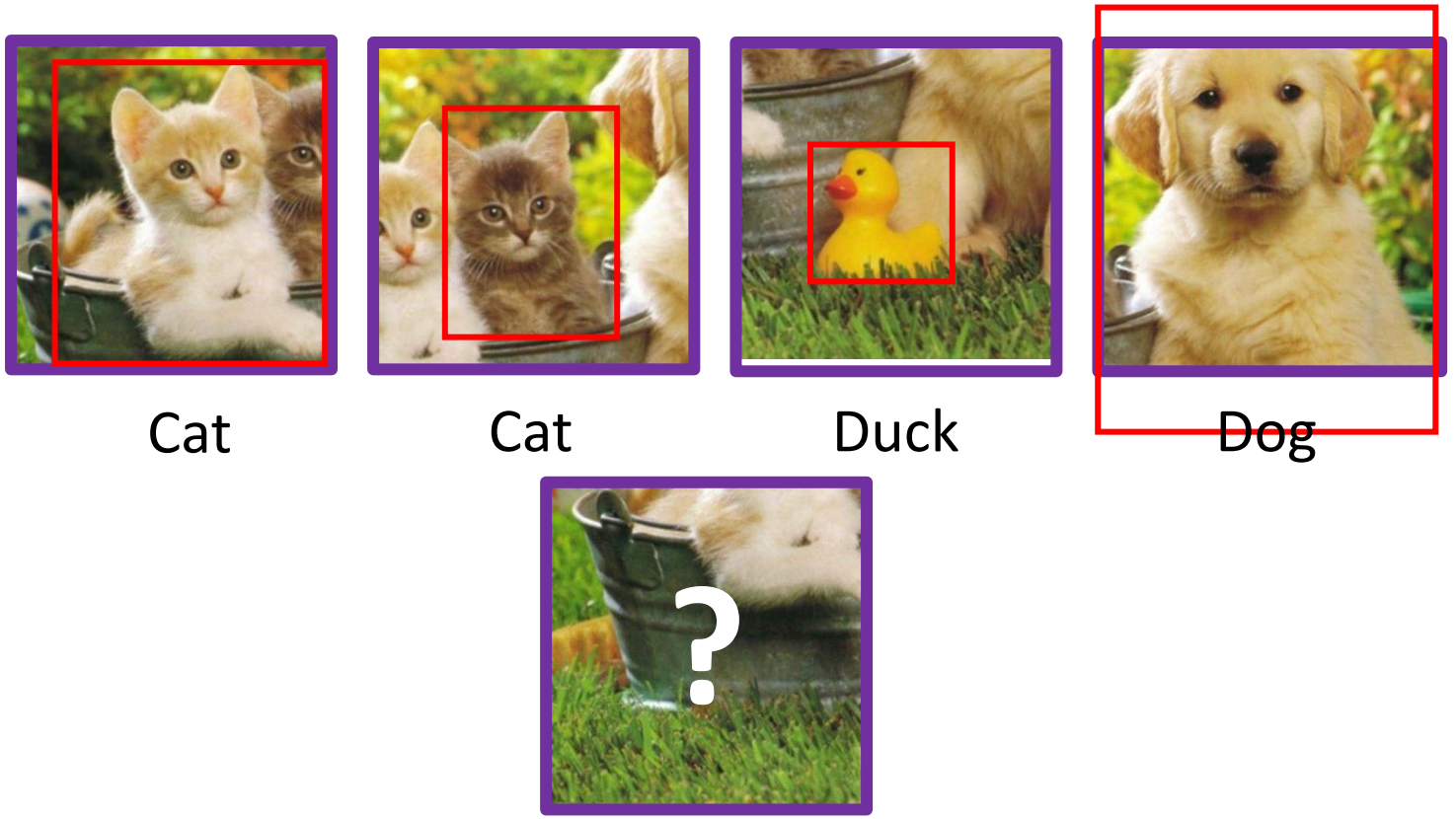
$\widehat{w}_{f\times 4}^{T} x$

$w_{f\times 1000}^{T} x$

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From Localization to Detection (v0)





“Background”

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From Localization to Detection (v0)



Deep net
("backbone")

$\widehat{w}_{f\times 4}^{T} x$

$w_{f\times 1000}^{T} x$

Challenges:

- Multiple types of outputs
- Number of objects

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From Localization to Detection (v0)



Deep net
("backbone")

$\widehat{w}_{f\times 4}^{T} x$

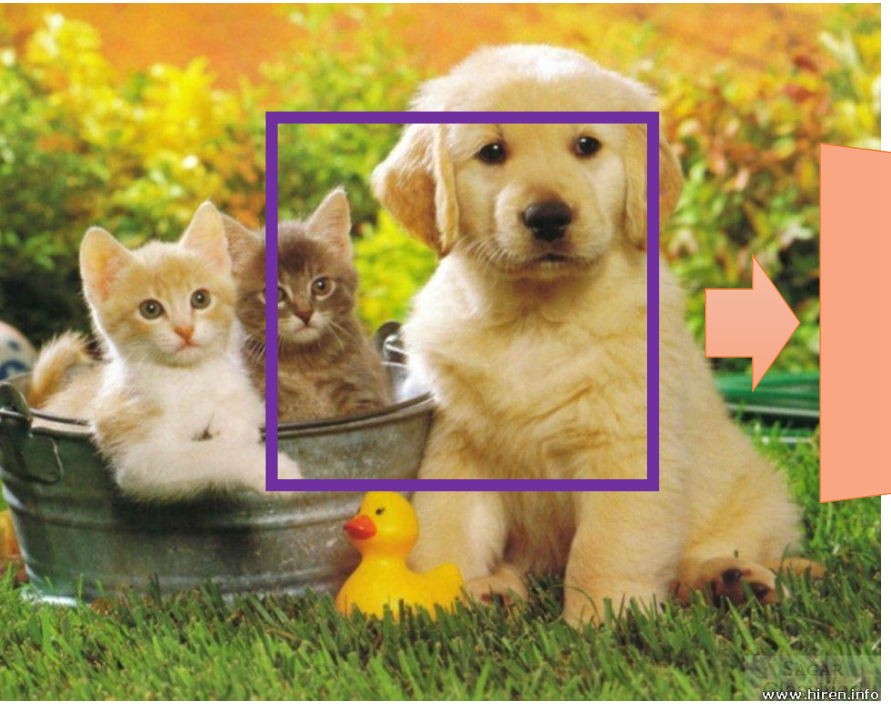
$w_{f\times 1000}^{T} x$

How many "sliding windows" are there?

There can easily be O(1M) windows!

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From Localization to Detection (v1)



Deep net
(“backbone”)

$$\hat{w}^{loc}_{f\times 1\times 1\times 4} * x$$

$$w^{class}_{f\times 1\times 1\times C} * x$$

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Object Detection

Single Shot:
SSD, YOLO…

Fast

High false rate



Two Shots:
R-CNN…

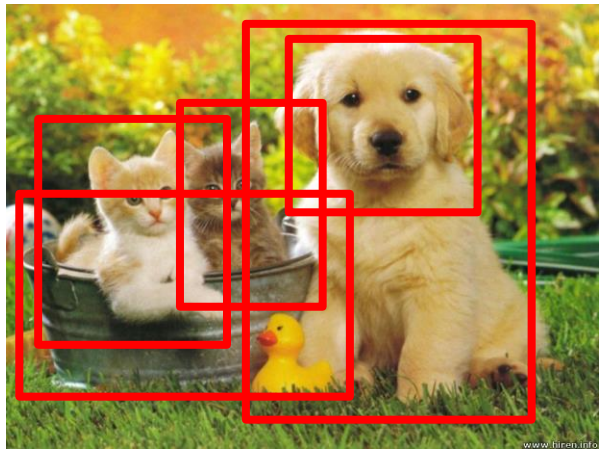
Slower

More accurate

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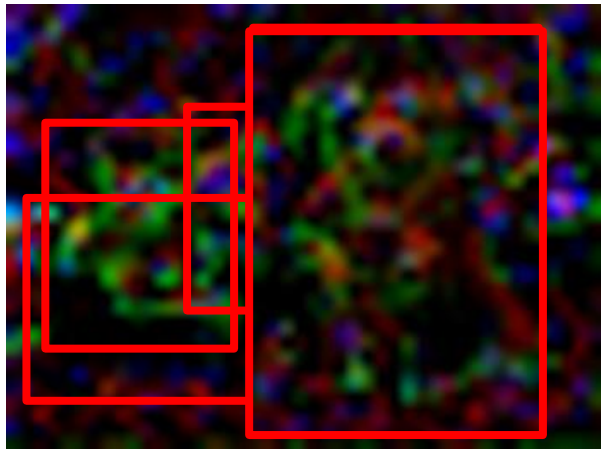
Faster R-CNN

RPN: Region Proposal Netwo



$3 \times H \times W$

backbone

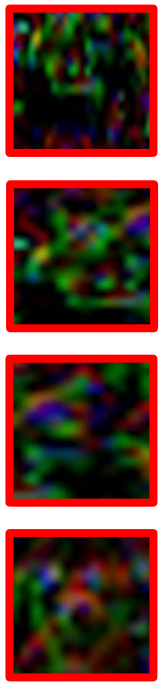


$D \times h \times w$

RPN

Class / BG

BBox

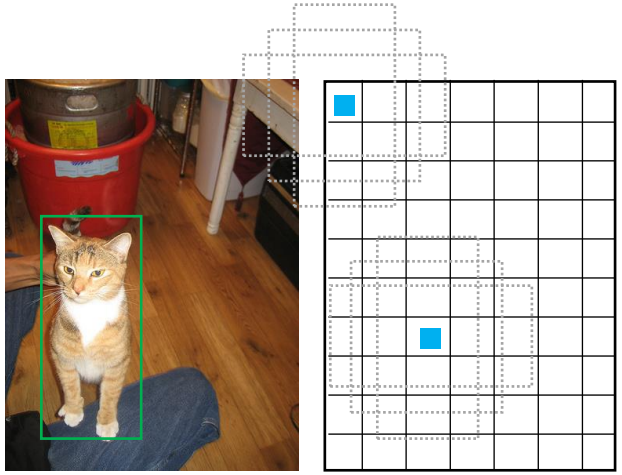


"ROI Pool" each proposed region from the **feature map**

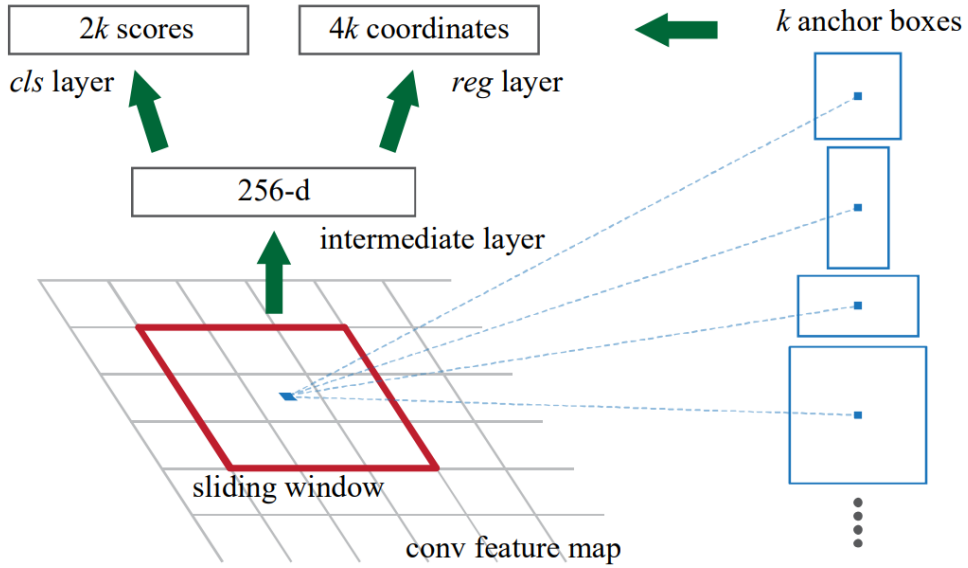
Ren, He, Girshick, and Sun "Faster r-cnn: Towards real-time object detection with region proposal networks" (NIPS 2015)

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RPN: Region Proposal Network



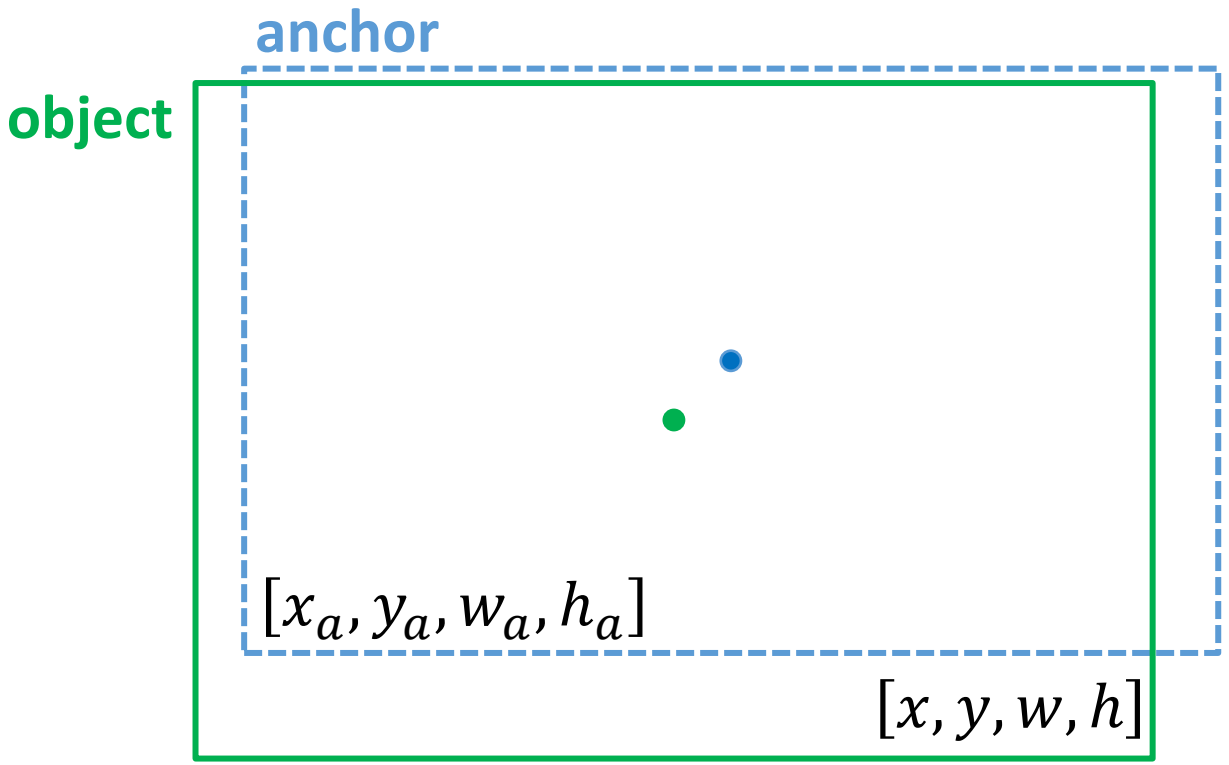
feature map



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RPN: Region Proposal Network

Predicting Bounding Box coordinates from anchors:

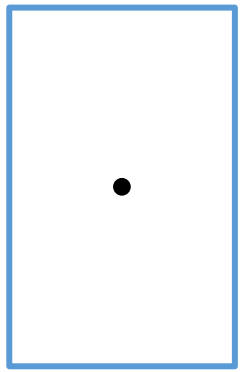


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RPN: Region Proposal Network

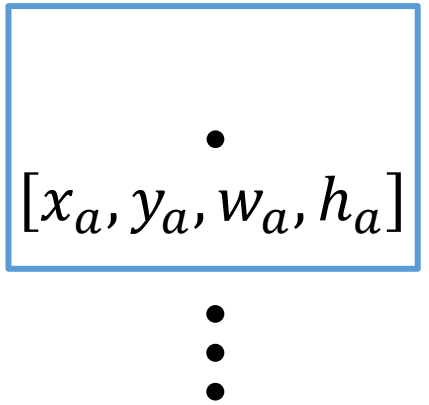
Predicting a bounding box "corrections" $\boldsymbol{t} = [t_x, t_y, t_w, t_h]$:

$$t_x = \frac{x - x_a}{w_a}, \qquad t_y = \frac{y - y_a}{h_a}, \qquad t_w = \log\left(\frac{w}{w_a}\right), \qquad t_h = \log\left(\frac{h}{h_a}\right)$$



Recovering the actual BBox from $\boldsymbol{t} = [t_x, t_y, t_w, t_h]$:

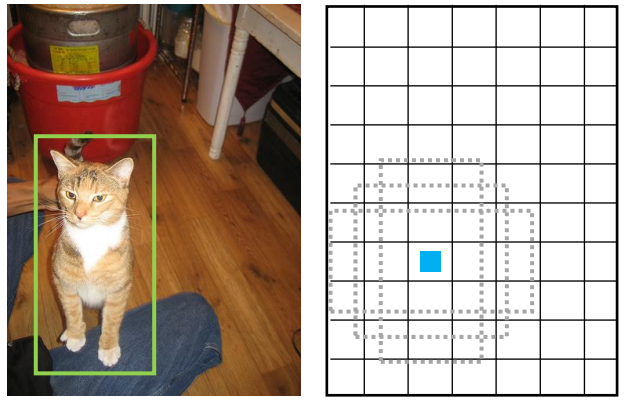
$$x = w_a \cdot t_x + x_a, \qquad y = h_a \cdot t_y + y_a,$$
$$w = e^{t_w} \cdot w_a, \qquad h = e^{t_h} \cdot h_a$$



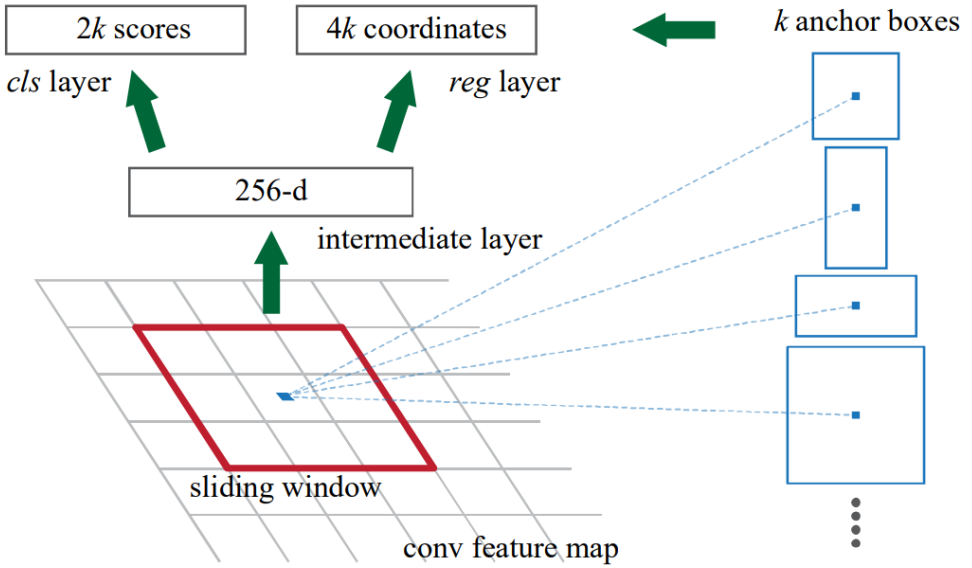
The learned "corrections" $\boldsymbol{t}$ are relative to anchor position and scale.

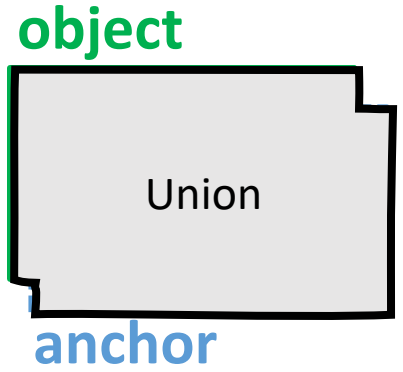
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Object Detection



Training



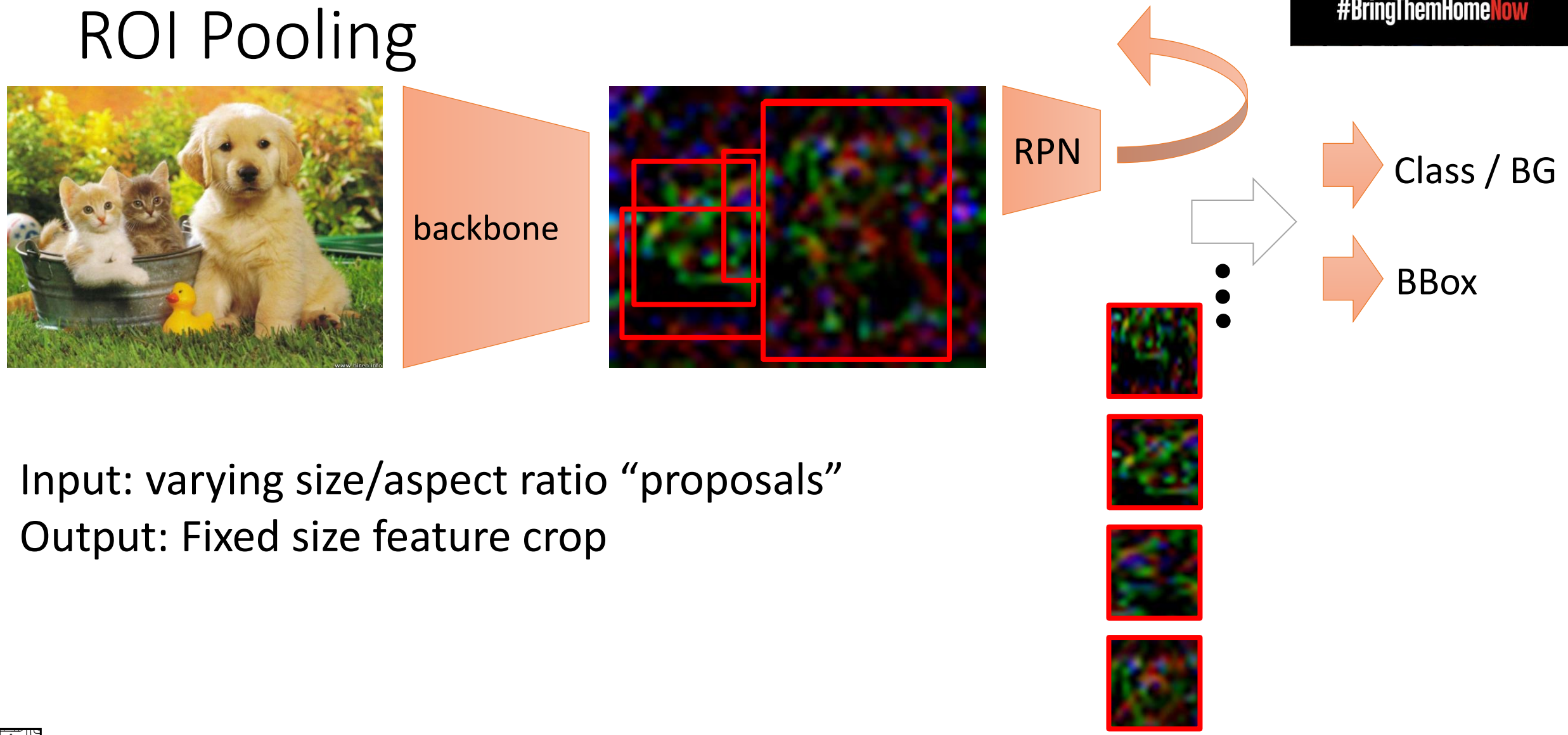


IoU = ——————

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ROI Pooling

RPN: Region **P**roposal **N**etwo

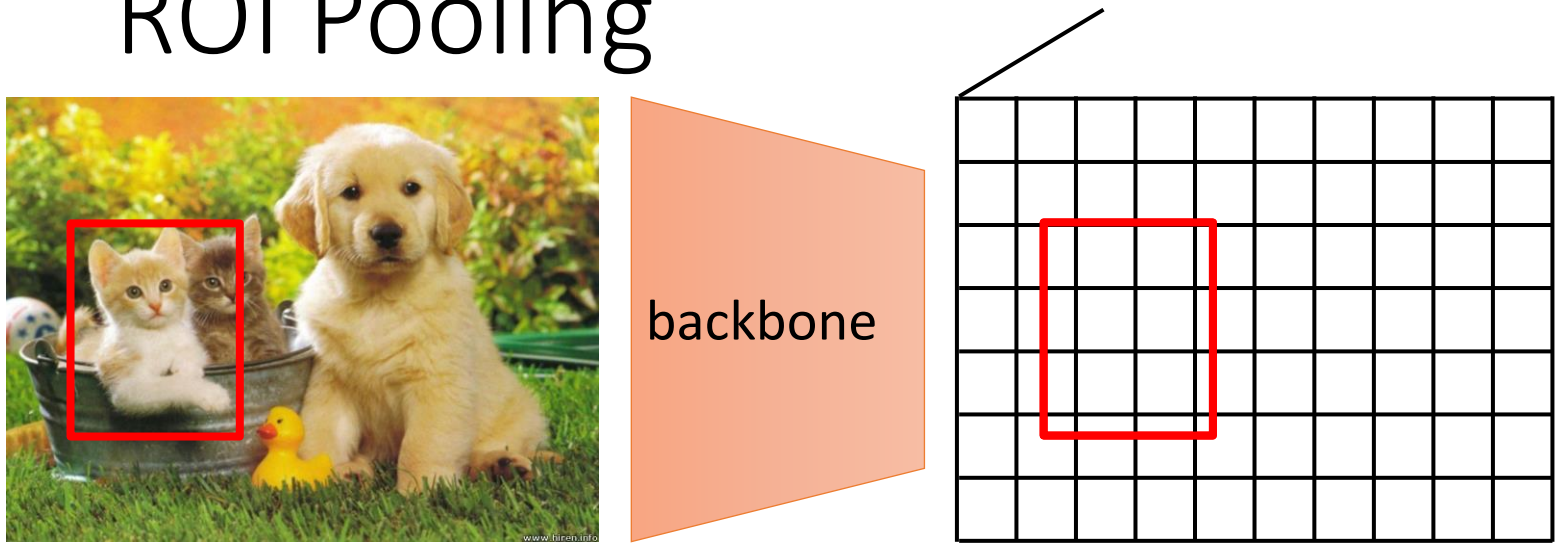


Input: varying size/aspect ratio “proposals”
Output: Fixed size feature crop

Ren, He, Girshick, and Sun "Faster r-cnn: Towards real-time object detection with region proposal networks" (NIPS 2015)

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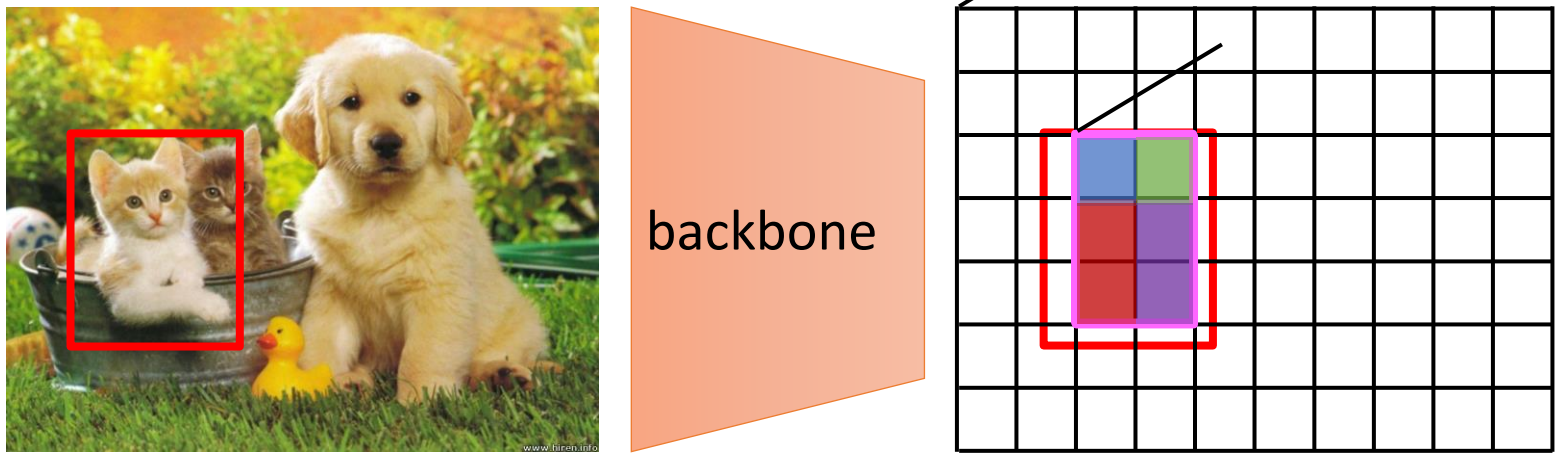
ROI Pooling



Input: varying size/aspect ratio “proposals”
Output: Fixed size feature crop

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ROI Pooling

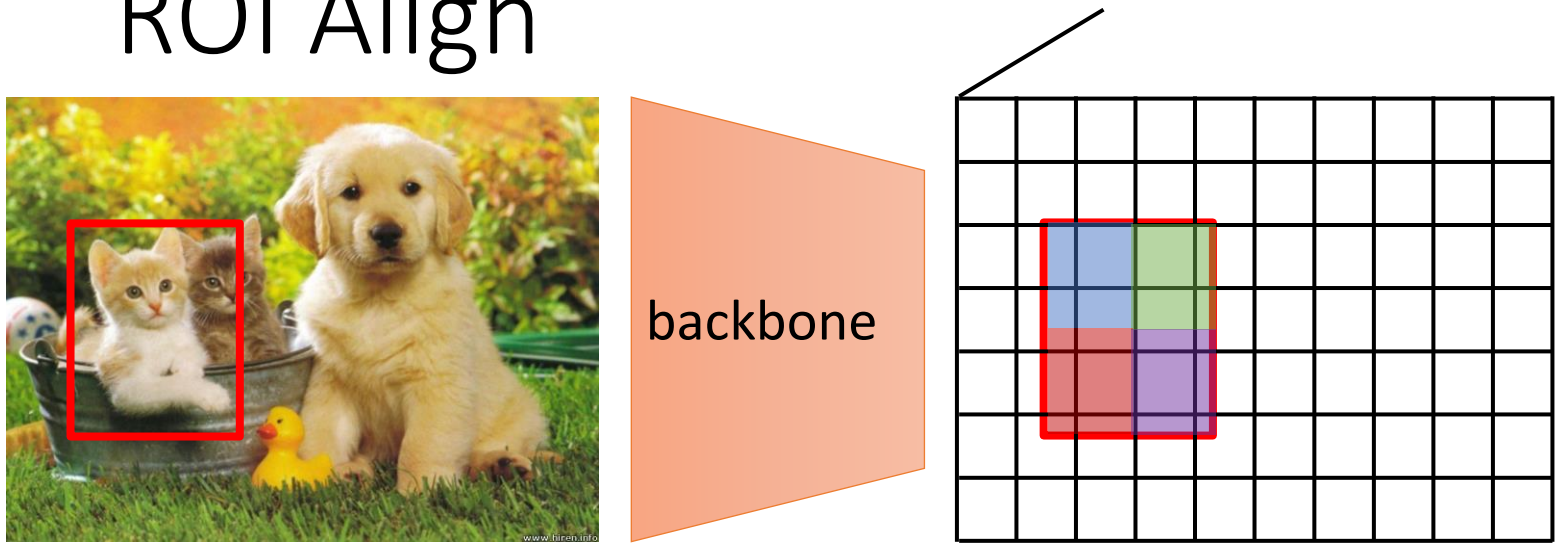


Input: varying size/aspect ratio “proposals”
Output: Fixed size feature crop

ROI Pool – “snap” to closest grid

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ROI Align



Input: varying size/aspect ratio “proposals”
Output: Fixed size feature crop

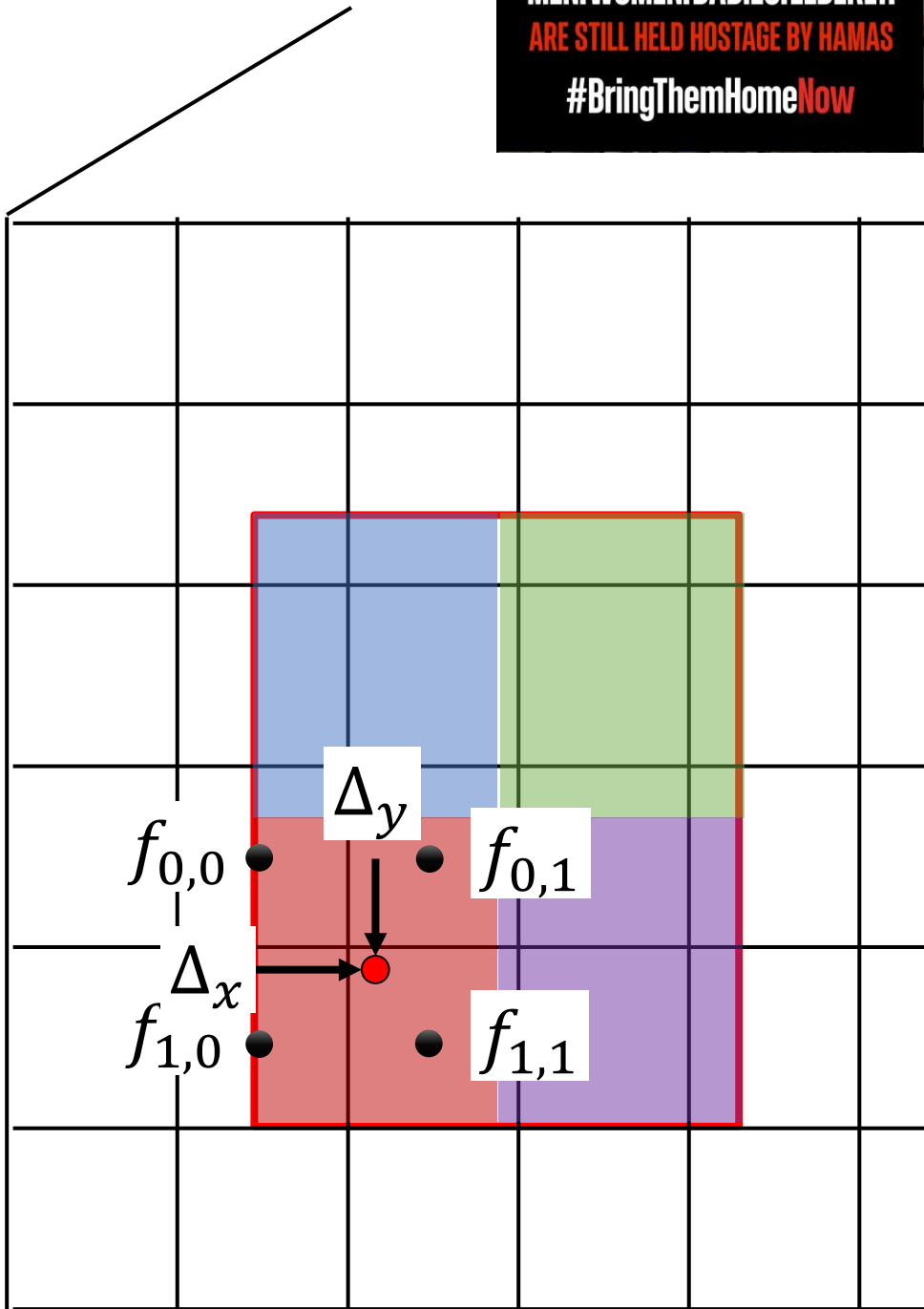
ROI Align – interp. features

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ROI Align

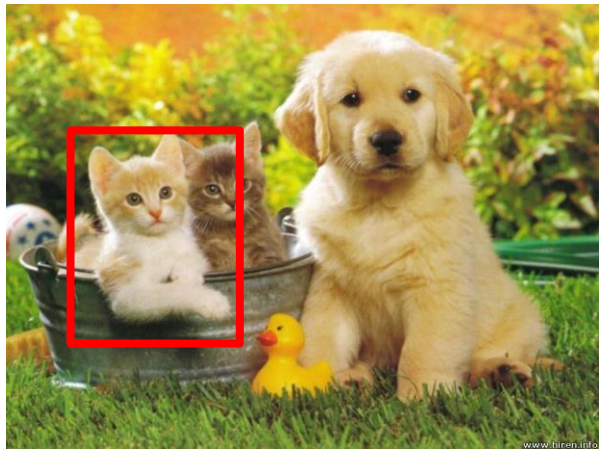
$$\boldsymbol{f}_{xy} = \sum_{ij} (1 - |x - x_i|)(1 - |y - y_i|)\boldsymbol{f}_{ij}$$

$$= (1 - \Delta_x)(1 - \Delta_y)\boldsymbol{f}_{0,0}$$
$$+\Delta_x(1 - \Delta_y)\boldsymbol{f}_{0,1}$$
$$+(1 - \Delta_x)\Delta_y\boldsymbol{f}_{1,0}$$
$$+\Delta_x\Delta_y\boldsymbol{f}_{1,1}$$

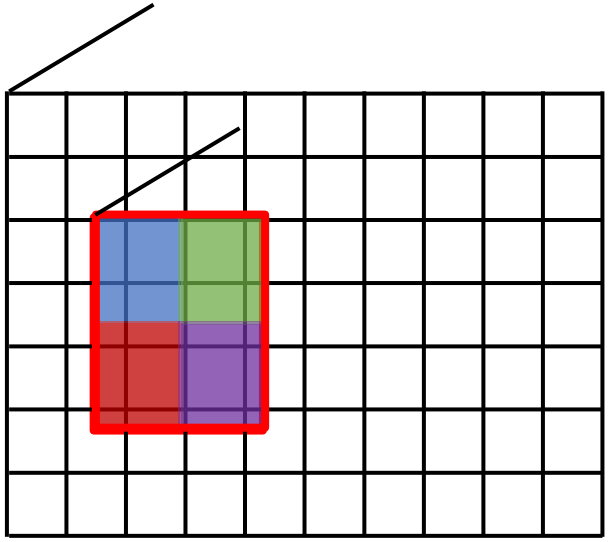


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ROI Align



backbone



Class / BG

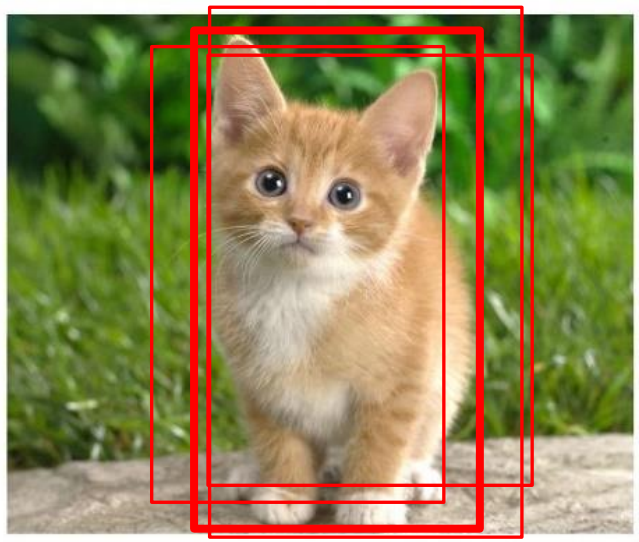
BBox

Input: varying size/aspect ratio “proposals”
Output: Fixed size feature crop

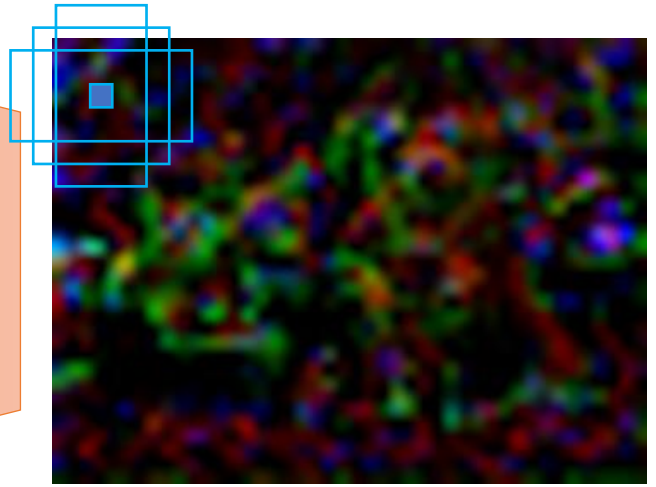
ROI Align – interp. features

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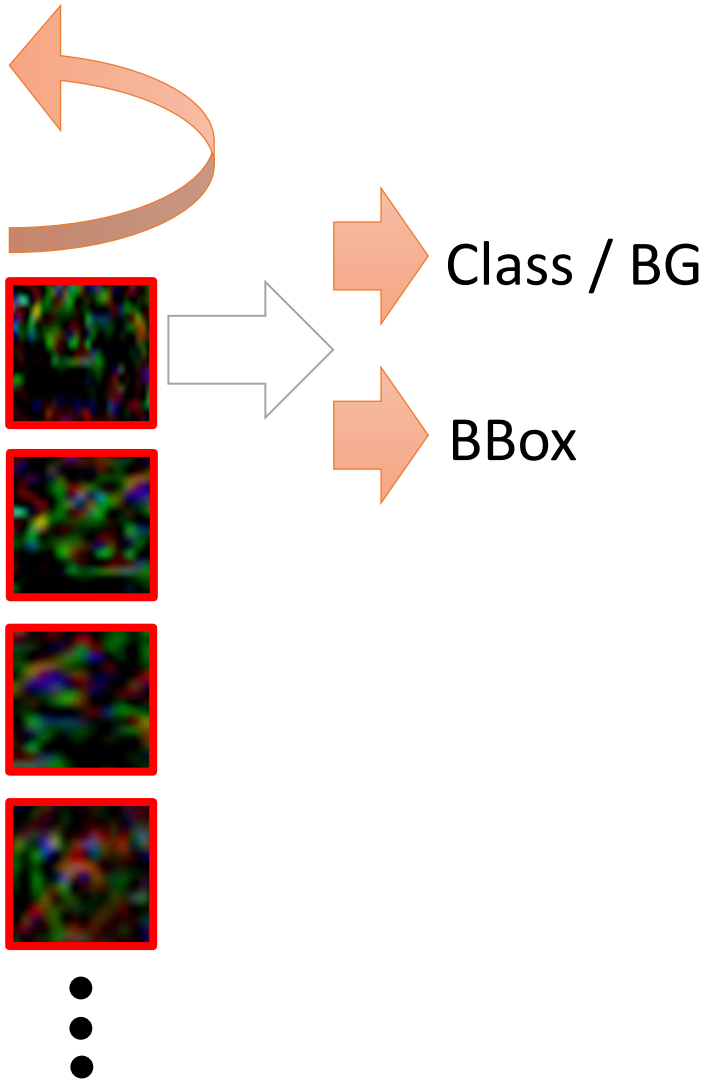
Inference



backbone



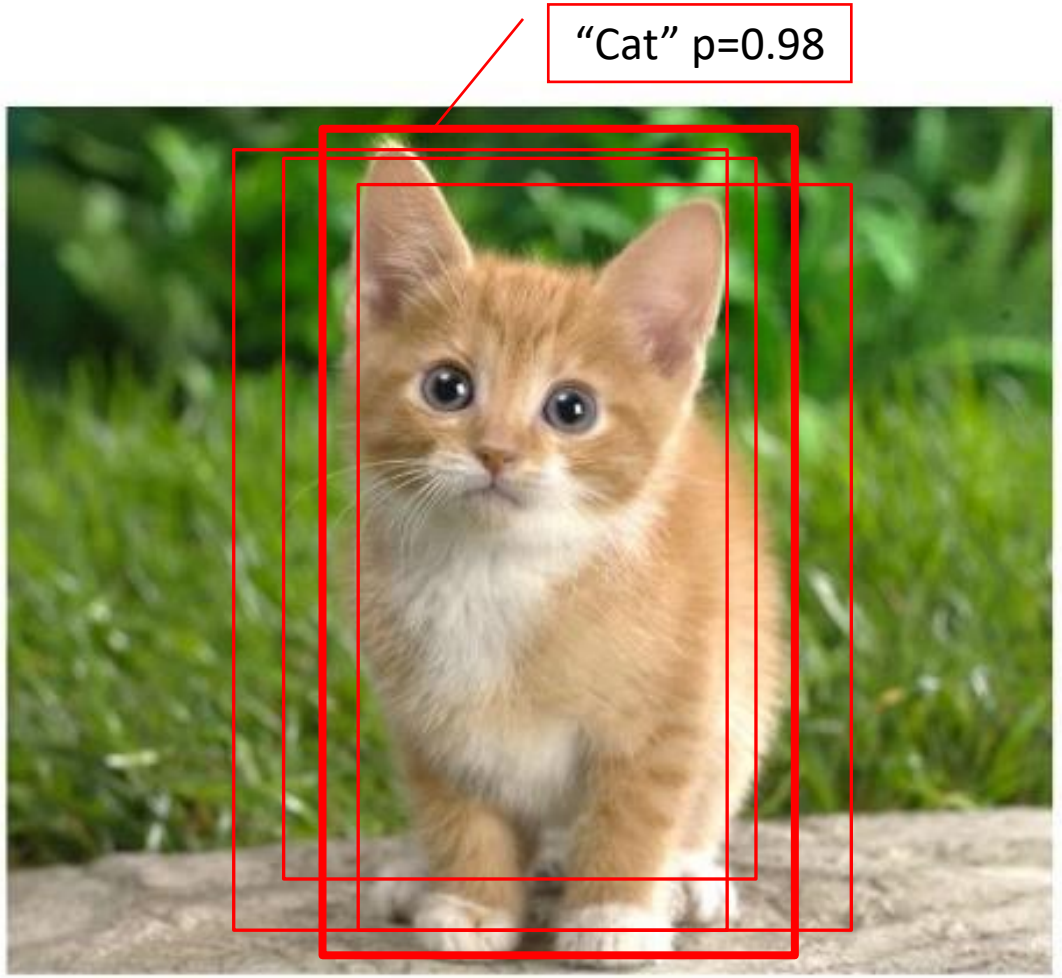
RPN



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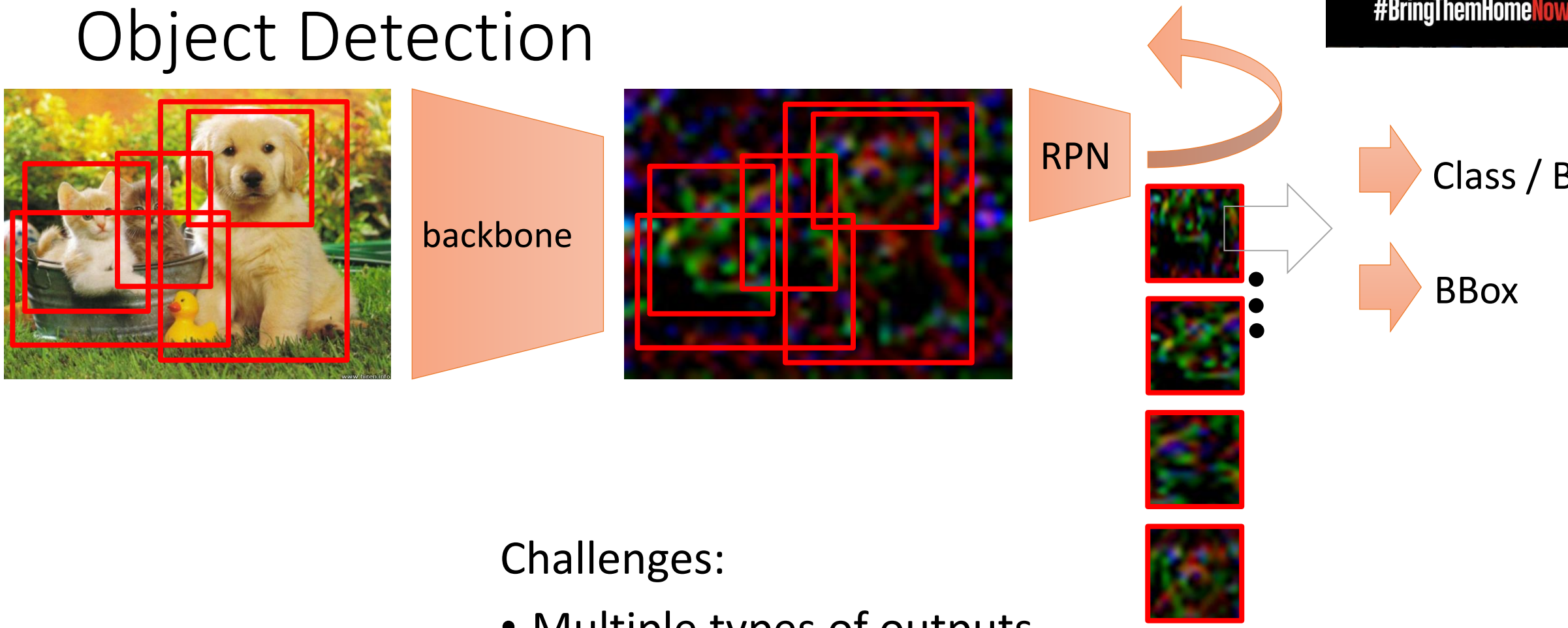
Inference: Non Maximal Suppression (NMS)

- Sort detection by score (per class)
- Take most confident
- Remove all overlapping
- Repeat



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Object Detection



Challenges:

- Multiple types of outputs
- Number of objects



Object Detection: Pitfalls

- Imbalance
- Multiscale

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Imbalance

Number of “negative” anchors ~O(10K)
Number of “positive” anchors ~O(10)

What happens to CE loss in this case?

$$CE(p_i, y_i) = -y_i \log(p_i)$$

$$= -9 \cdot \log(0.5) - 9 \cdot \log(0.5)$$

$$=$$

Prediction p_i \ True y_i	BG	Obj
BG	9981	9
Obj	9	1

Unconfident: *p=0.5*

Confident: *p=0.99*

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Imbalance

Number of “negative” anchors ~O(10K)
Number of “positive” anchors ~O(10)

What happens to CE loss in this case?

$$\nabla CE(p_i, y_i) = p_i - y_i$$

$$= 9 \cdot 0.5 + 9 \cdot (1 - 0.5)$$

Prediction p_i \ True y_i	BG	Obj
BG	9981	9
Obj	9	1

Unconfident: *p=0.5*

Confident: *p=0.99*

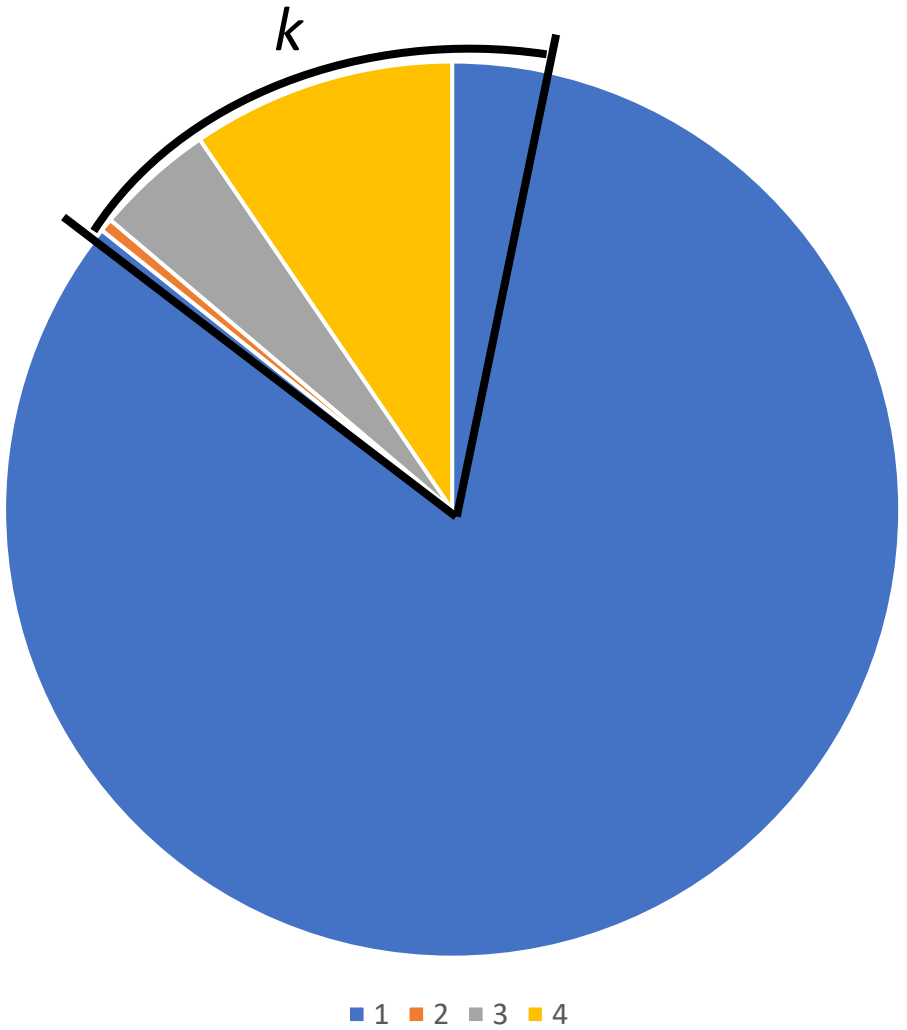
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Imbalance – Hard Negative Mining

Compute loss for all *N* anchors

Select top *k* “hard” examples

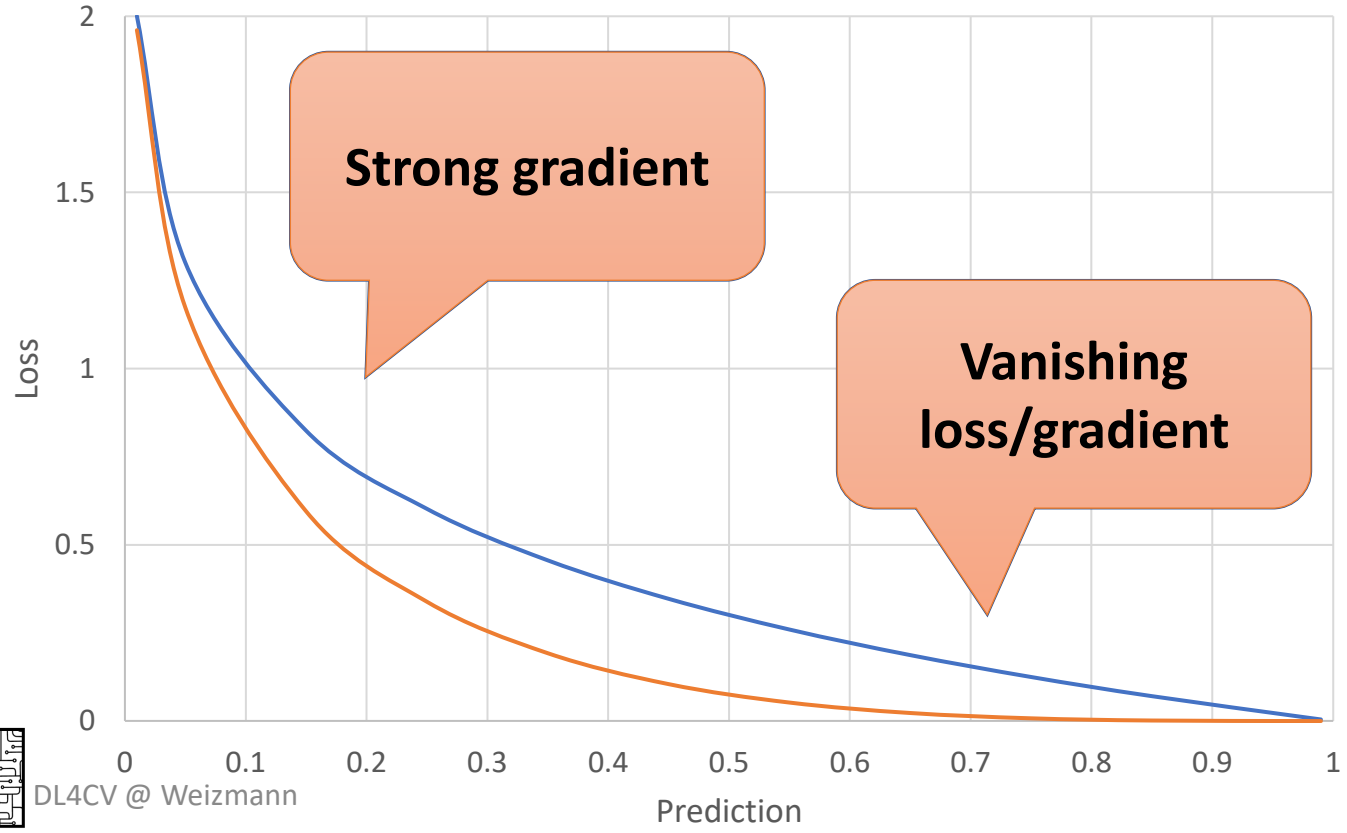
Compute gradient for hard *k* ***only***



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Imbalance – Focal Loss

Lin,Goyal, Girshick, He, and Dollár
Focal loss for dense object detection (PAMI 2018)



$$CE = -y_i \log p_i$$
$$FL = -y_i(\mathbf{1} - \boldsymbol{p_i})^{\gamma} \log p_i$$



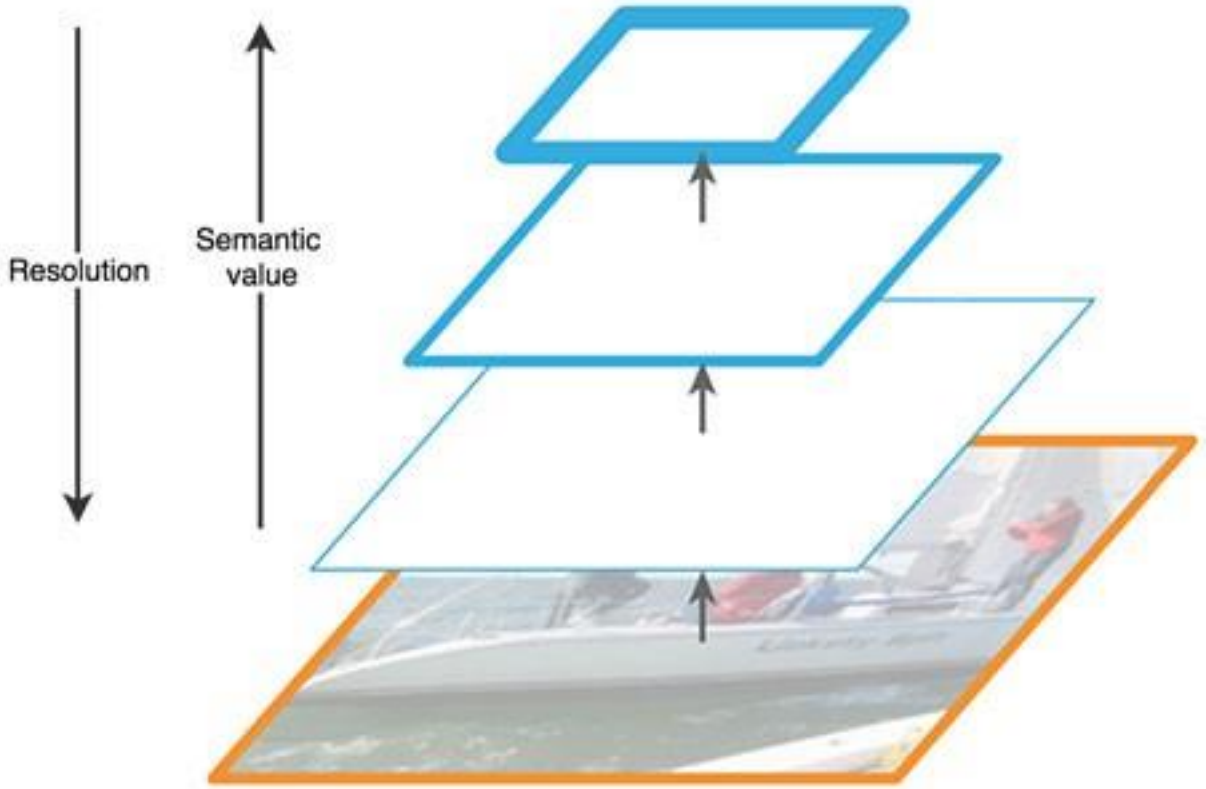
Object Detection: Pitfalls

- Imbalance
- Multiscale

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Resolution vs Semantic Value

How to handle multiscale predictions?

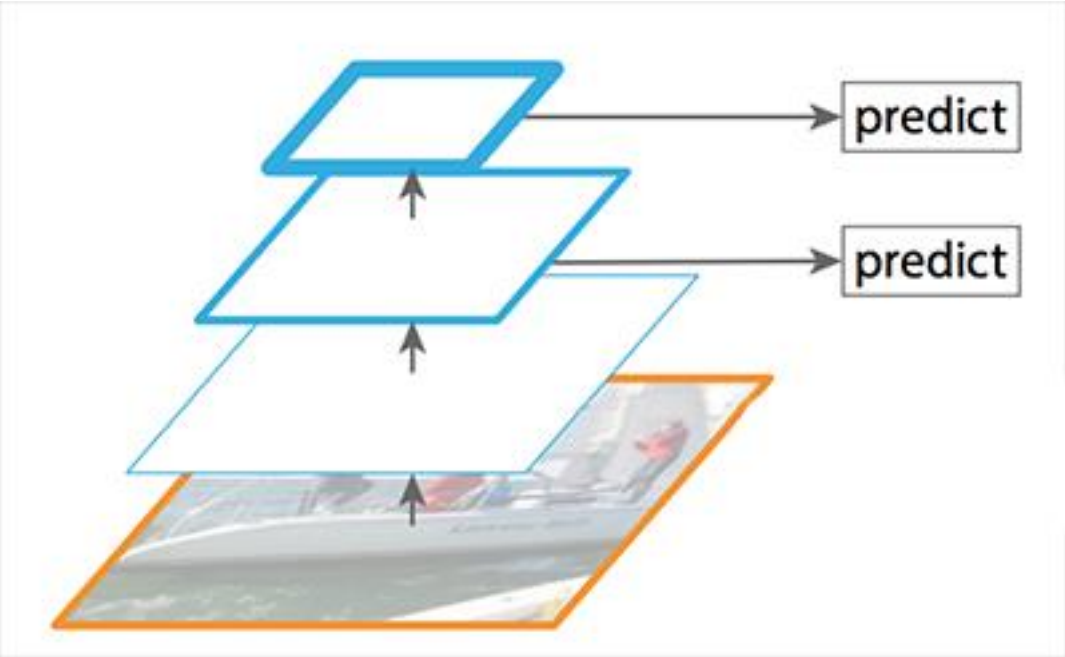


Tsung-Yi, Dollár, Girshick, He, Hariharan and Belongie. Feature Pyramid Networks for Object Detection (CVPR 2017)

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Resolution vs Semantic Value

How to handle multiscale predictions?

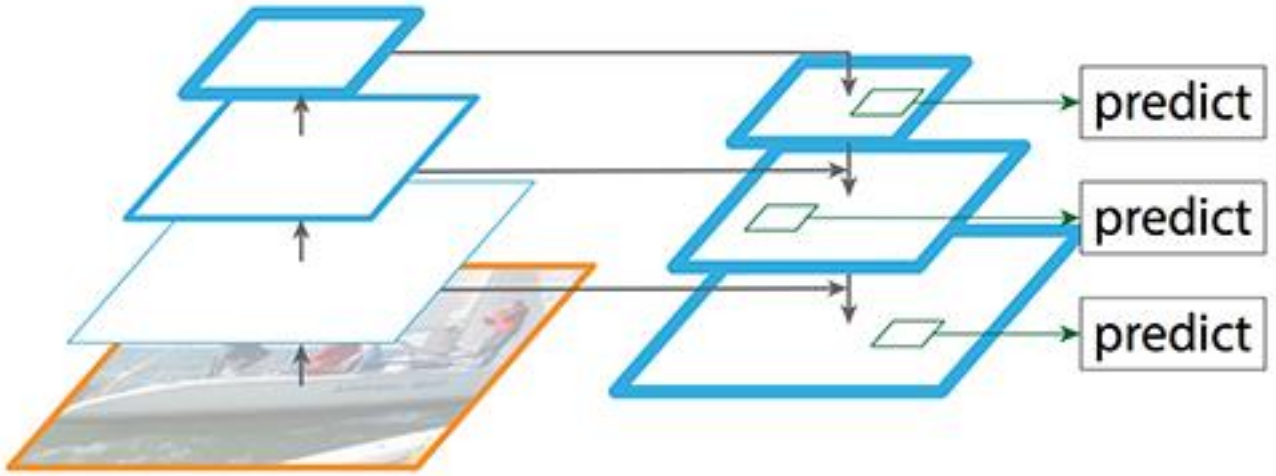


Tsung-Yi, Dollár, Girshick, He, Hariharan and Belongie. Feature Pyramid Networks for Object Detection (CVPR 2017)

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Feature Pyramid Network (FPN)

How to handle multiscale predictions?

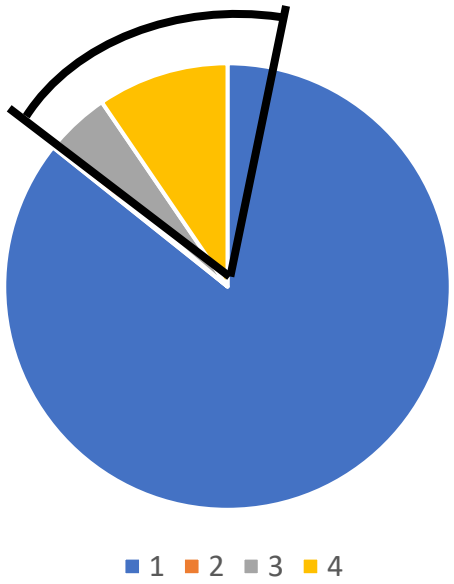


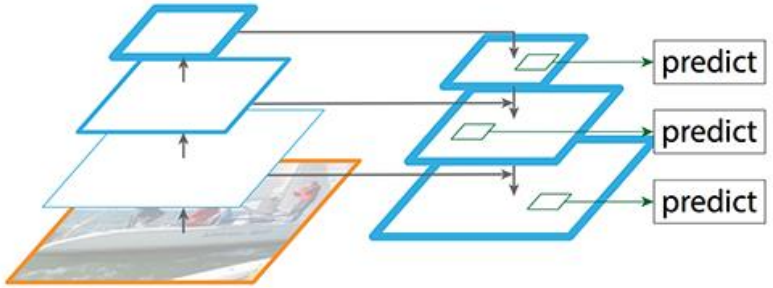
Tsung-Yi, Dollár, Girshick, He, Hariharan and Belongie. **Feature Pyramid Networks for Object Detection** (CVPR 2017)

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Object Detection

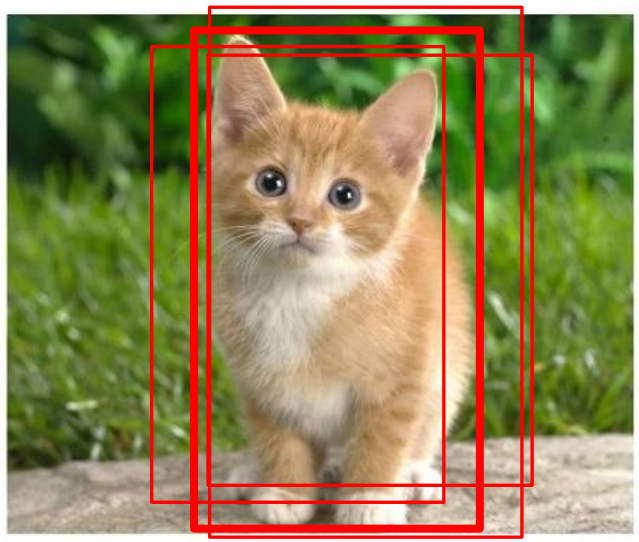
- Imbalance data
- Multiscale



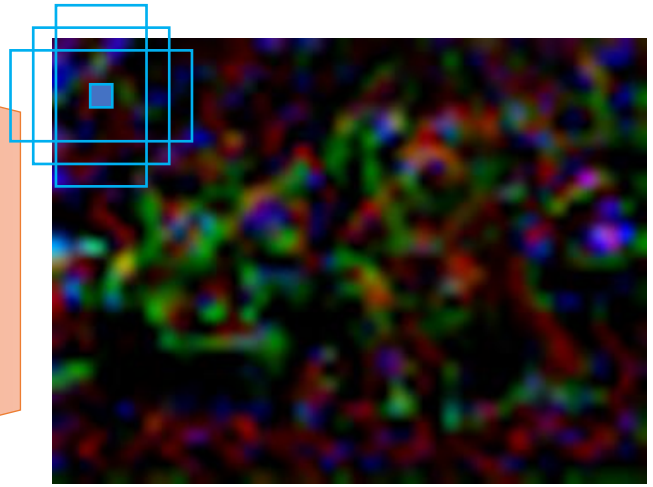


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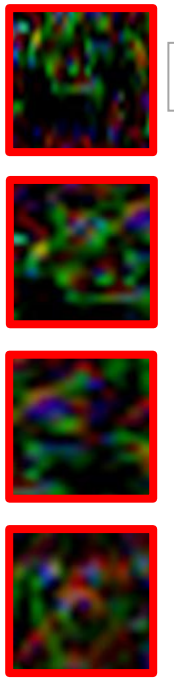
Object Detection



backbone



RPN



Class / BG

BBox

Can we do better?

“Set” prediction

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Semantic Segmentation



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Semantic Segmentation



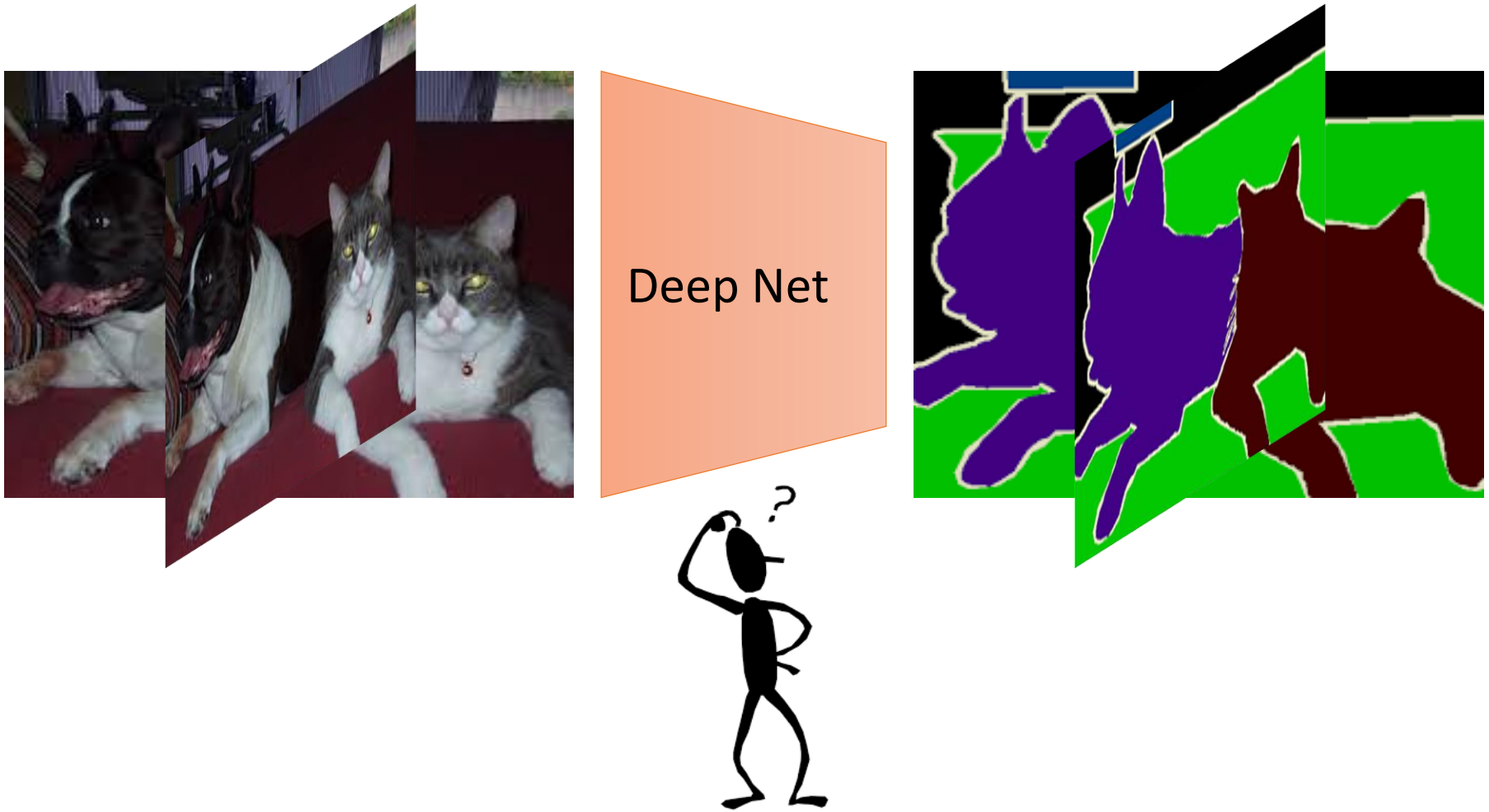
Deep Net





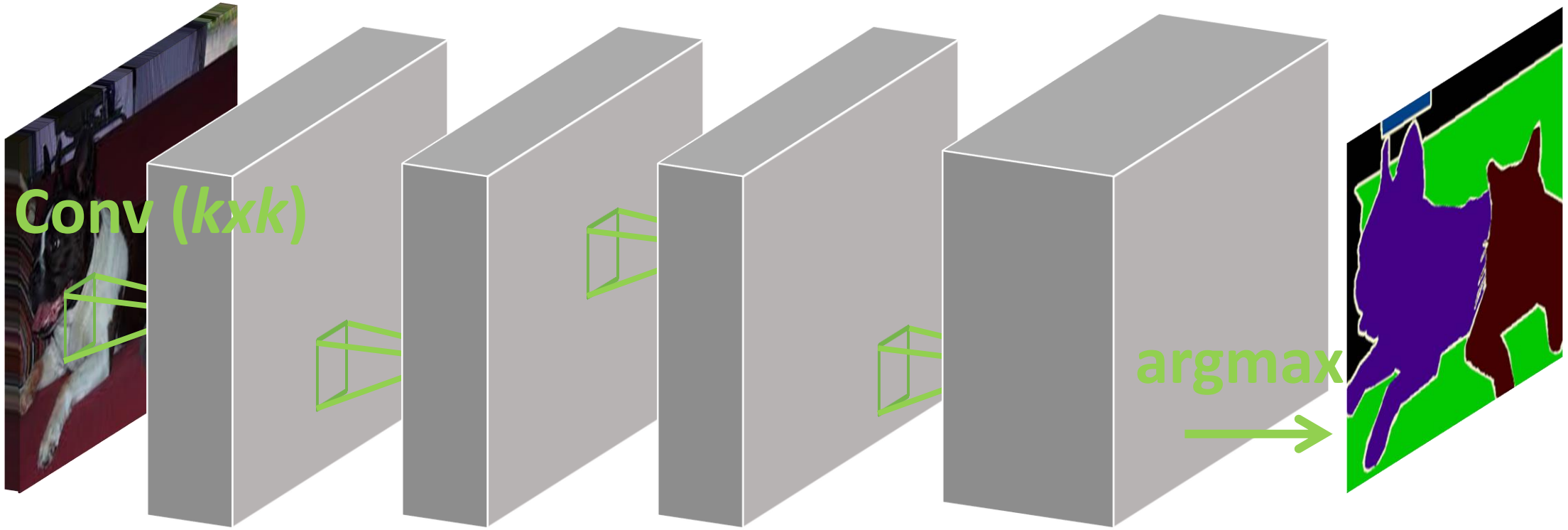
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Semantic Segmentation



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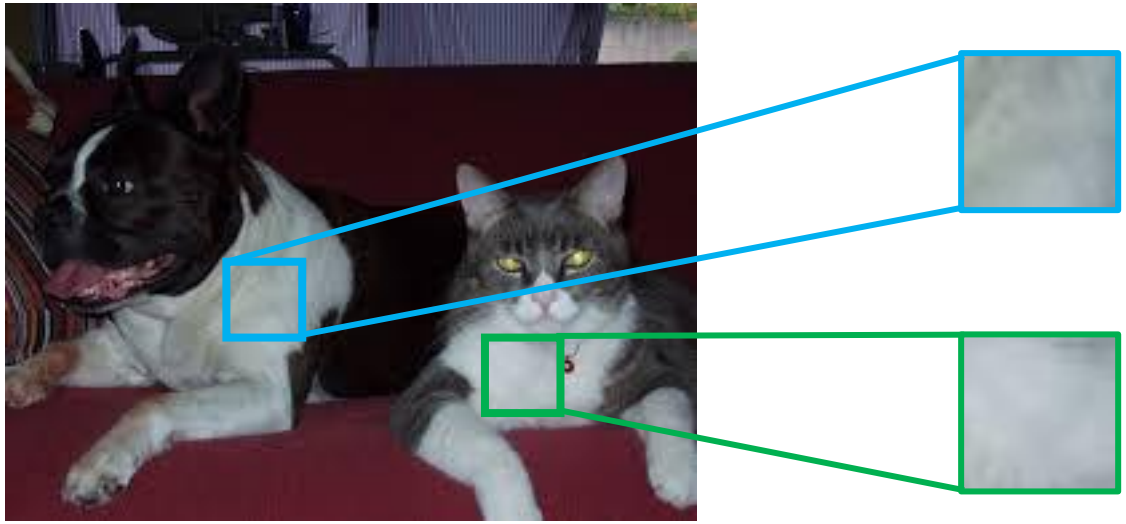
Semantic Segmentation (v0)



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Semantic Segmentation (v0)

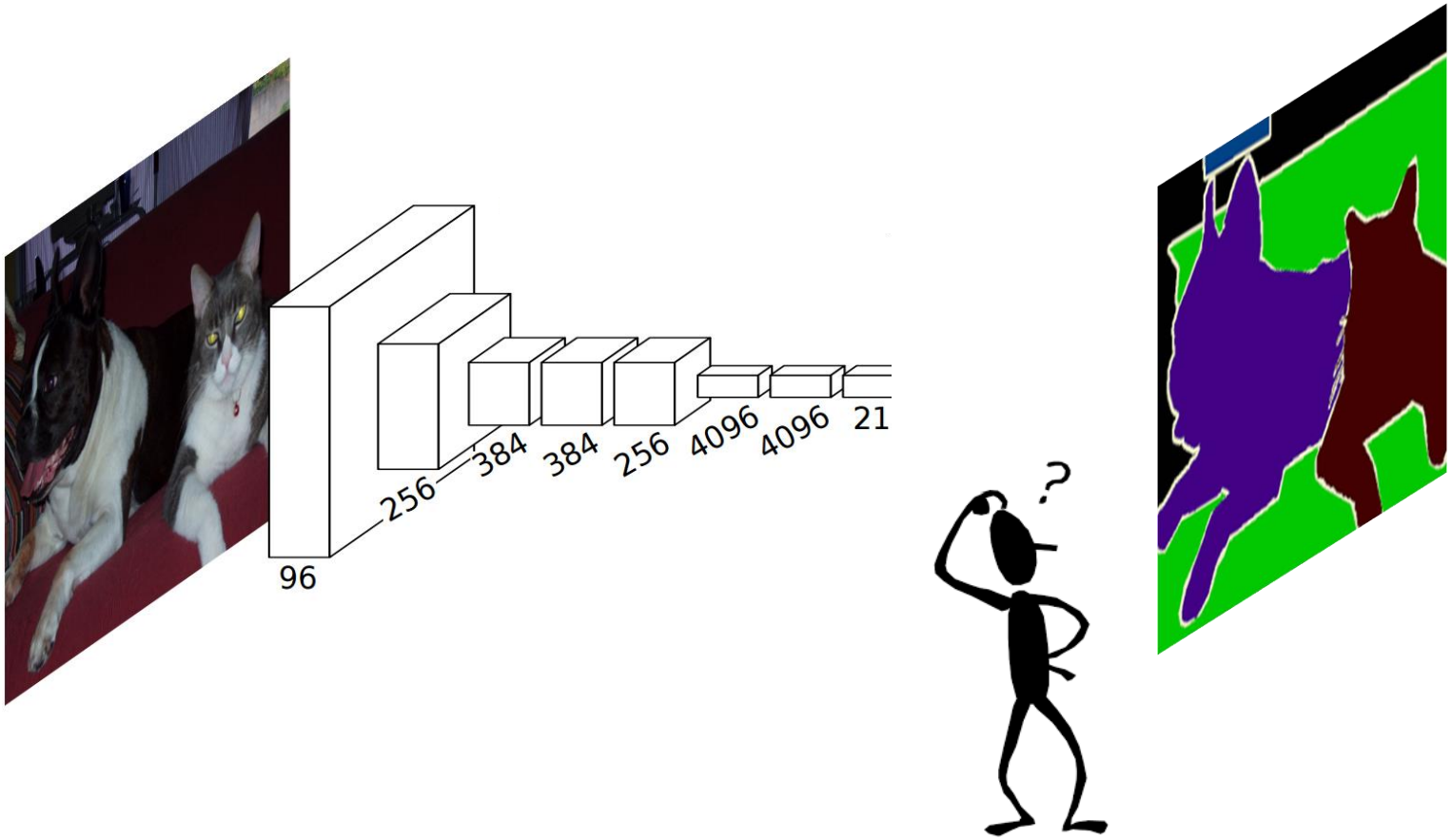
Challenge: Context (RF) vs Resolution



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Semantic Segmentation - FCN

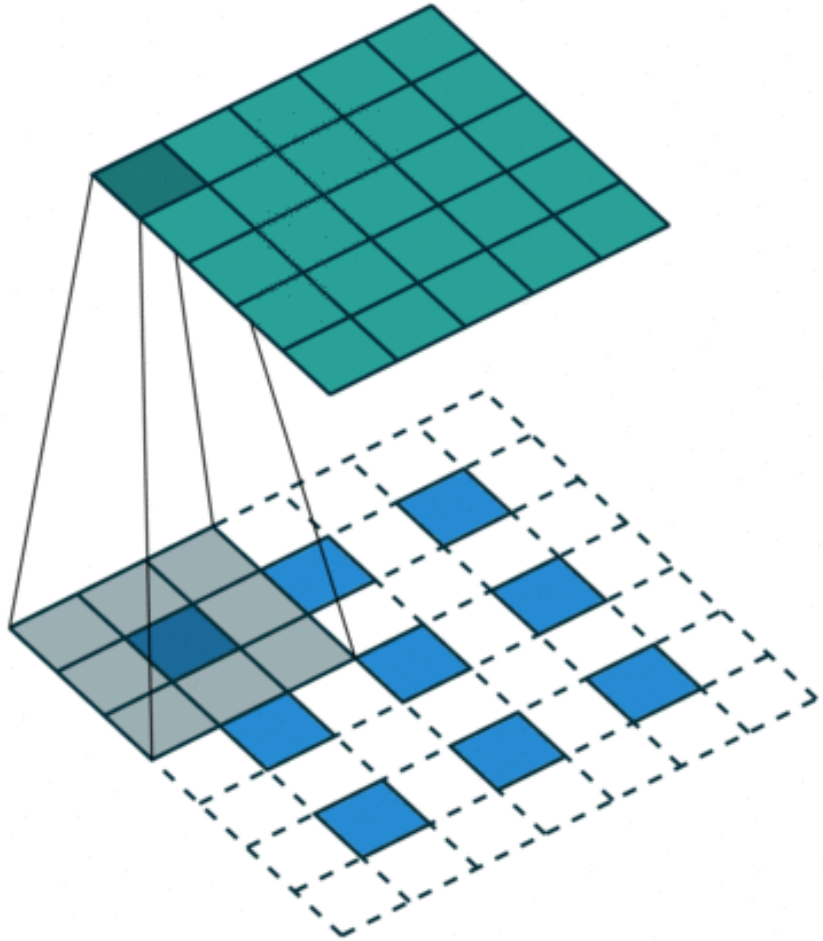
Replace FC layers with conv – “sliding window” classification



Long, Shelhamer and Darrell Fully convolutional networks for semantic segmentation (CVPR 2015)

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Transposed Convolution

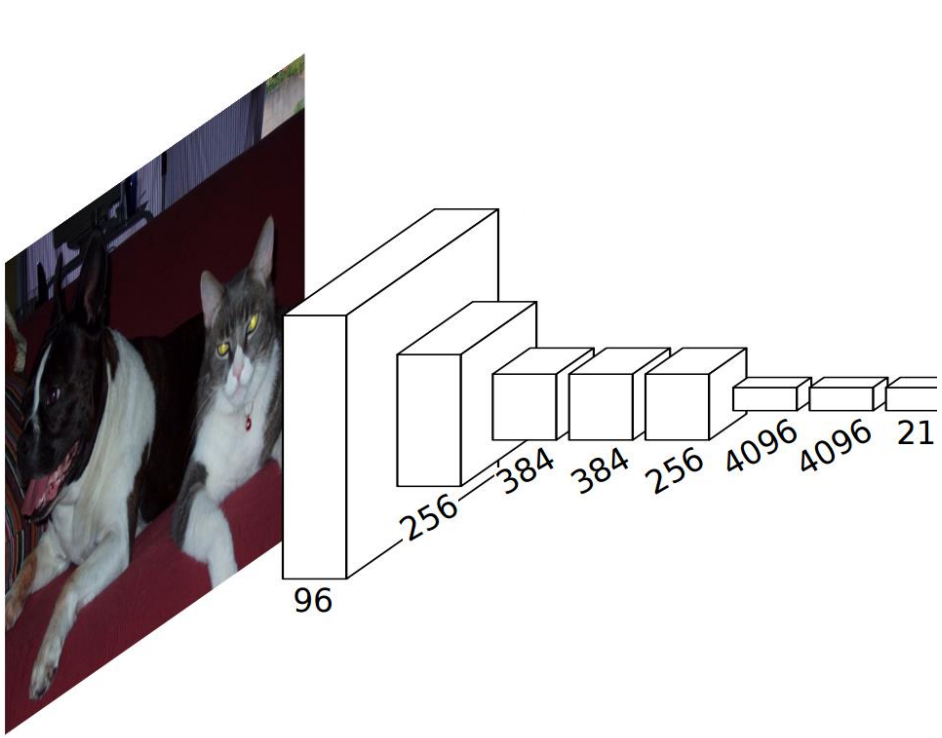


In depth: her

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Semantic Segmentation - FCN

Replace FC layers with conv – “sliding window” classification



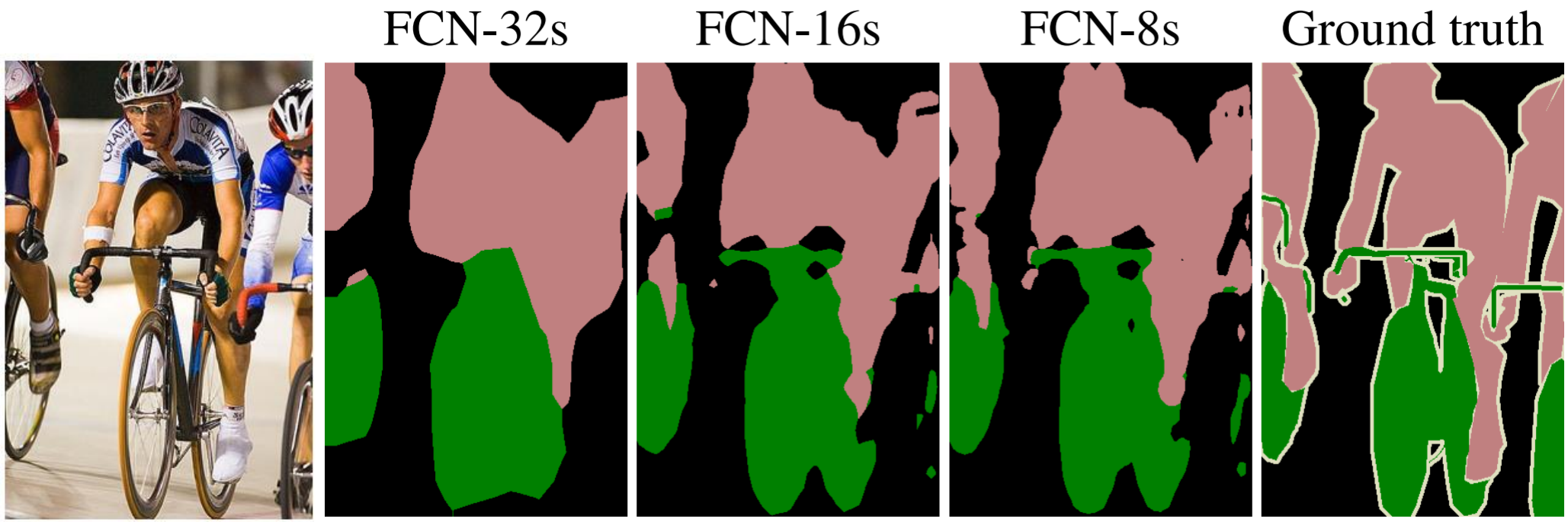




Long, Shelhamer and Darrell [Fully convolutional networks for semantic segmentation](#) (CVPR 2015)

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Semantic Segmentation - FCN

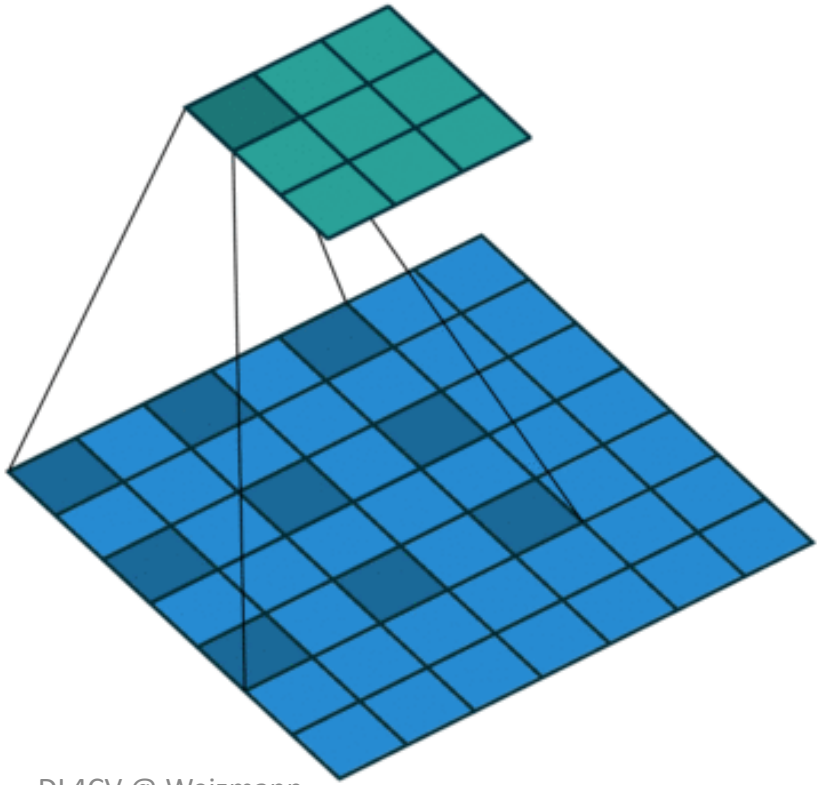


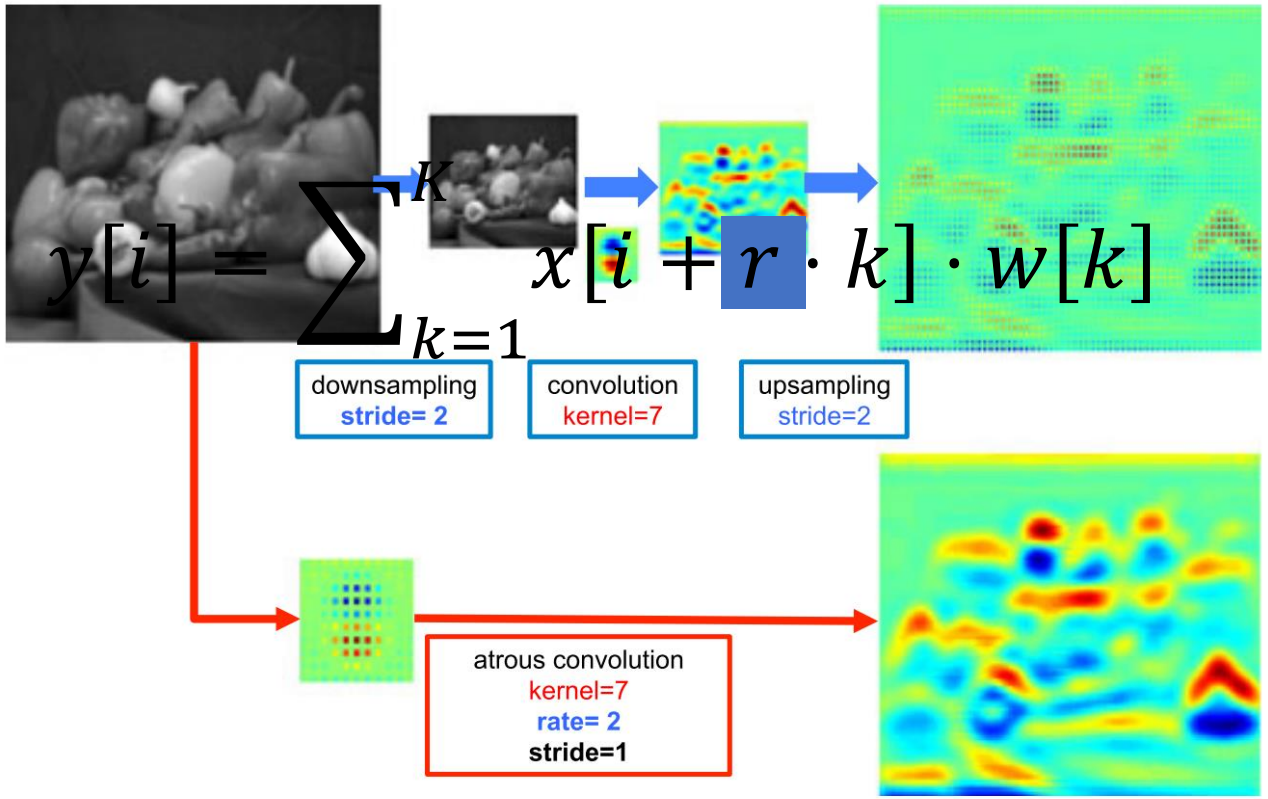
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DeepLab: Atrous Convolution

Chen, Papandreou, Kokkinos, Murphy and Yuille

Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs (*PAMI 2018*)



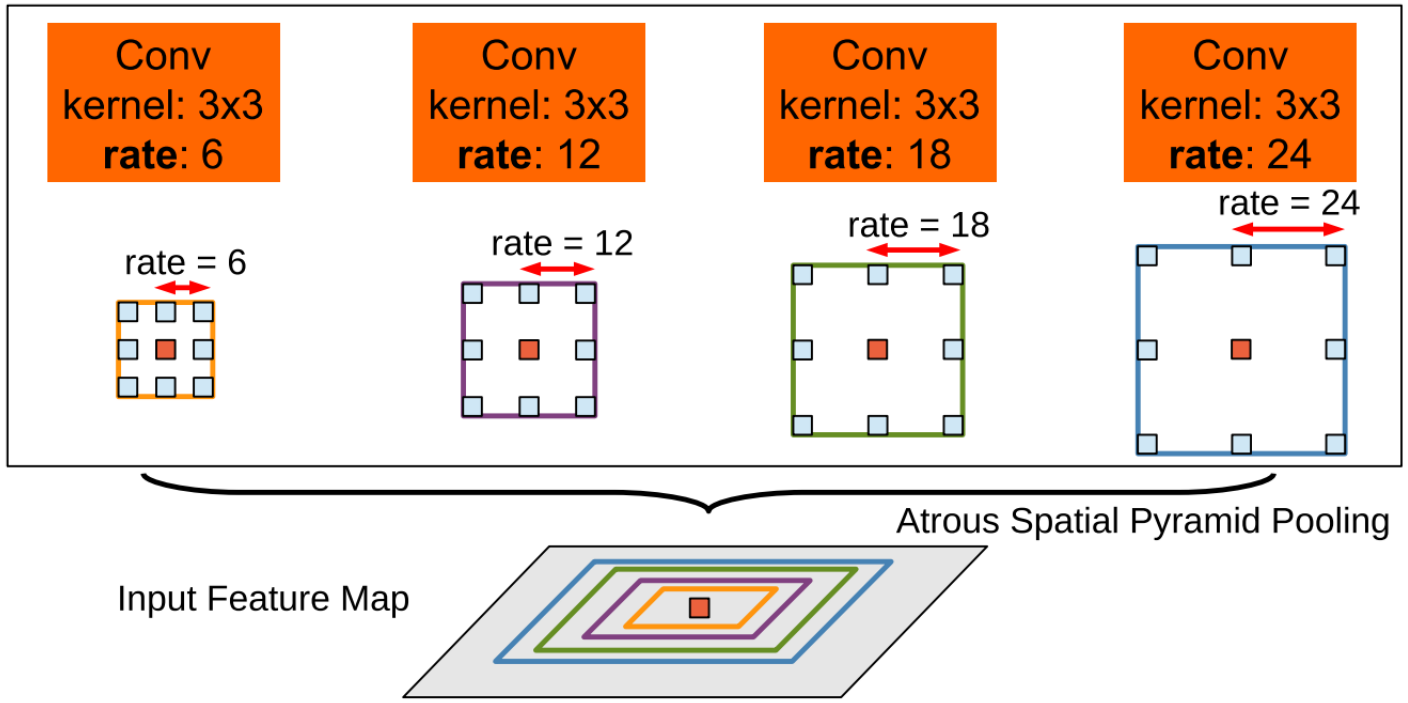


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DeepLab: Atrous Convolution

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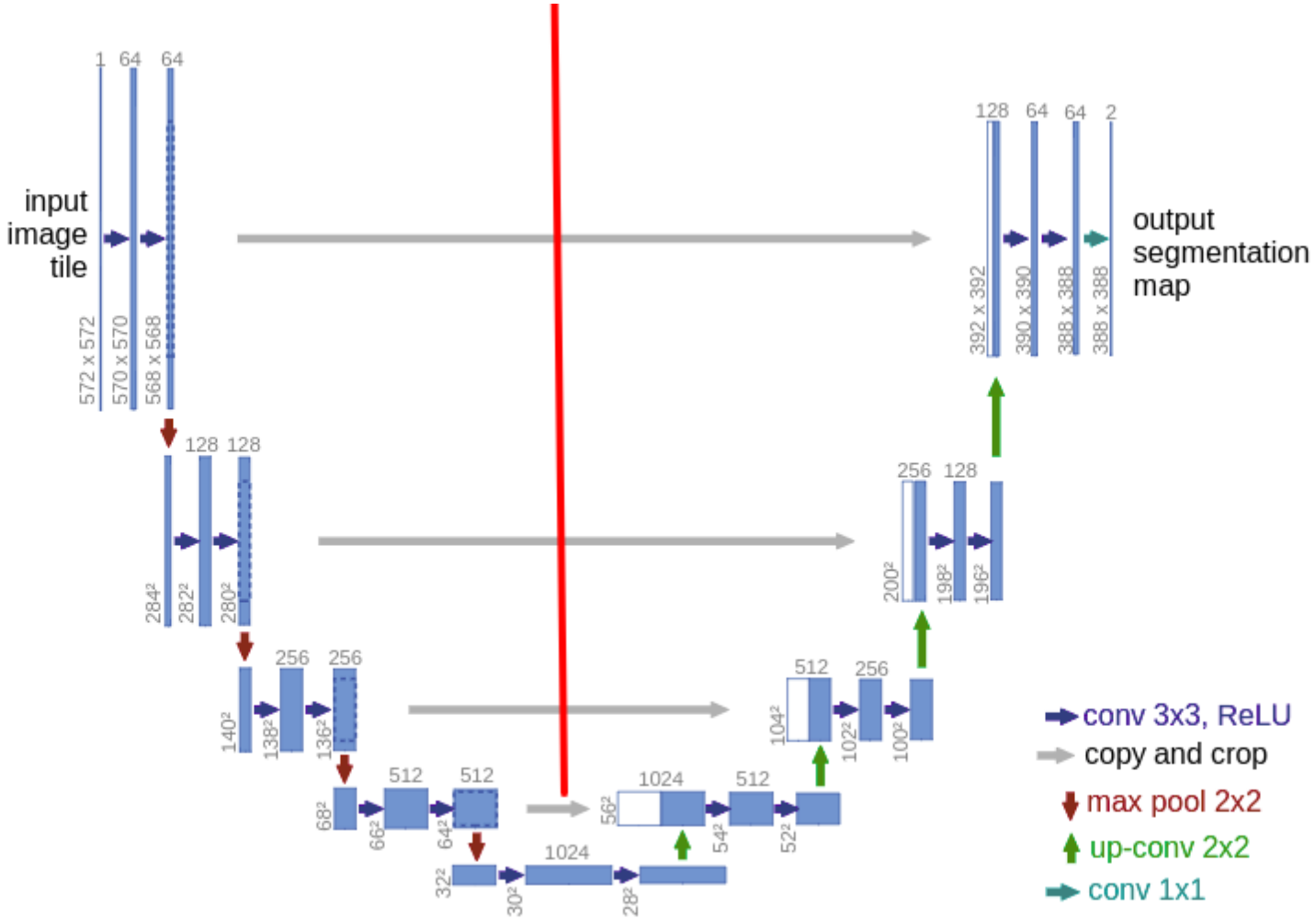
DeepLab: Atrous Convolution

- Trade stride/pooling with “dilation” of kernel
- Increase receptive field without increase in parameters



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Semantic Segmentation – U-net

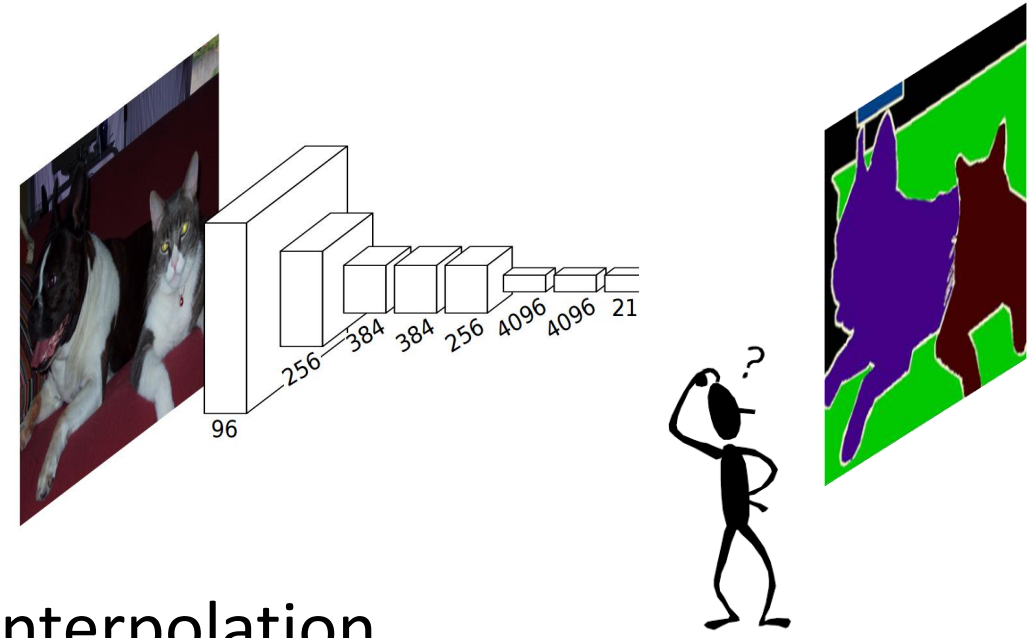


Olaf, Fischer and Brox U-net: Convolutional networks for biomedical image segmentation (2015)

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Semantic Segmentation

Resolution vs. Semantic information



- FCN: using “transposed convolution”
- DeepLab: dilated convolution + simple interpolation
- U-net: skip connections

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Instance Segmentation



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Instance Segmentation: Detection or Segmentation?

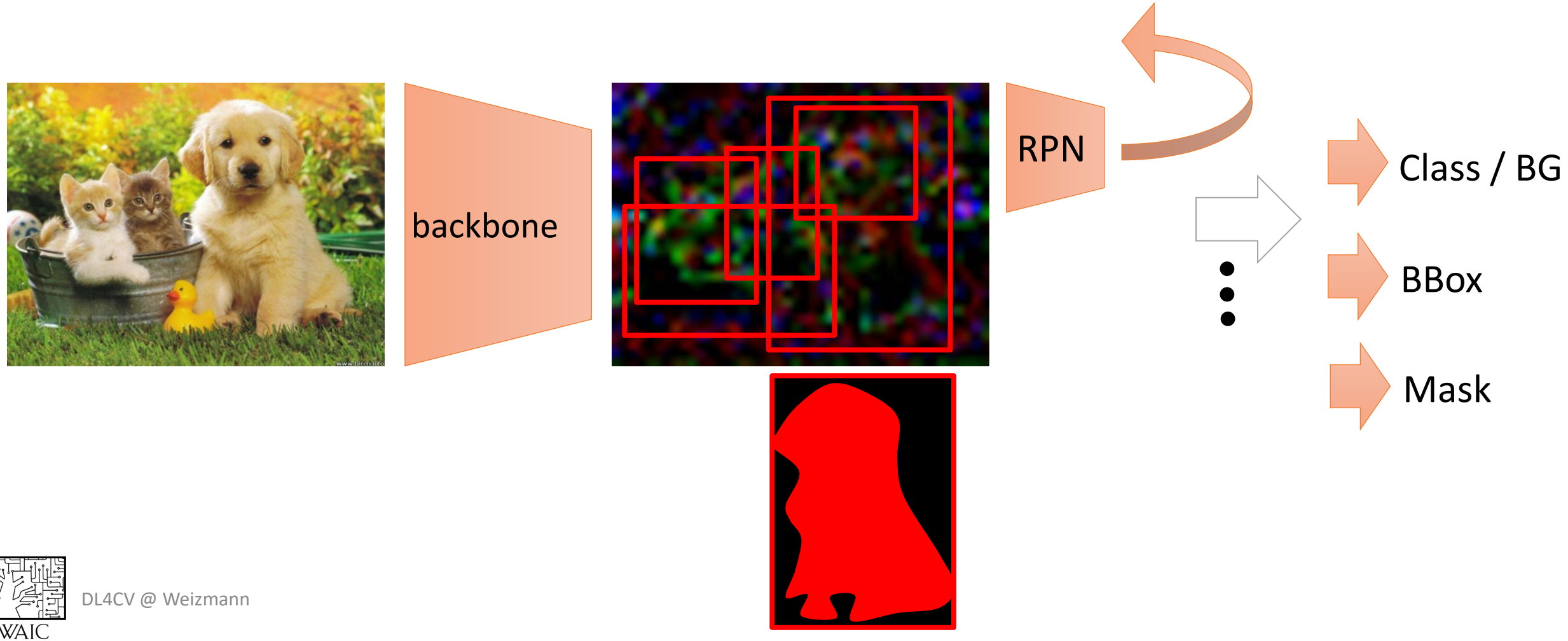




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Instance Segmentation: Mask R-CNN

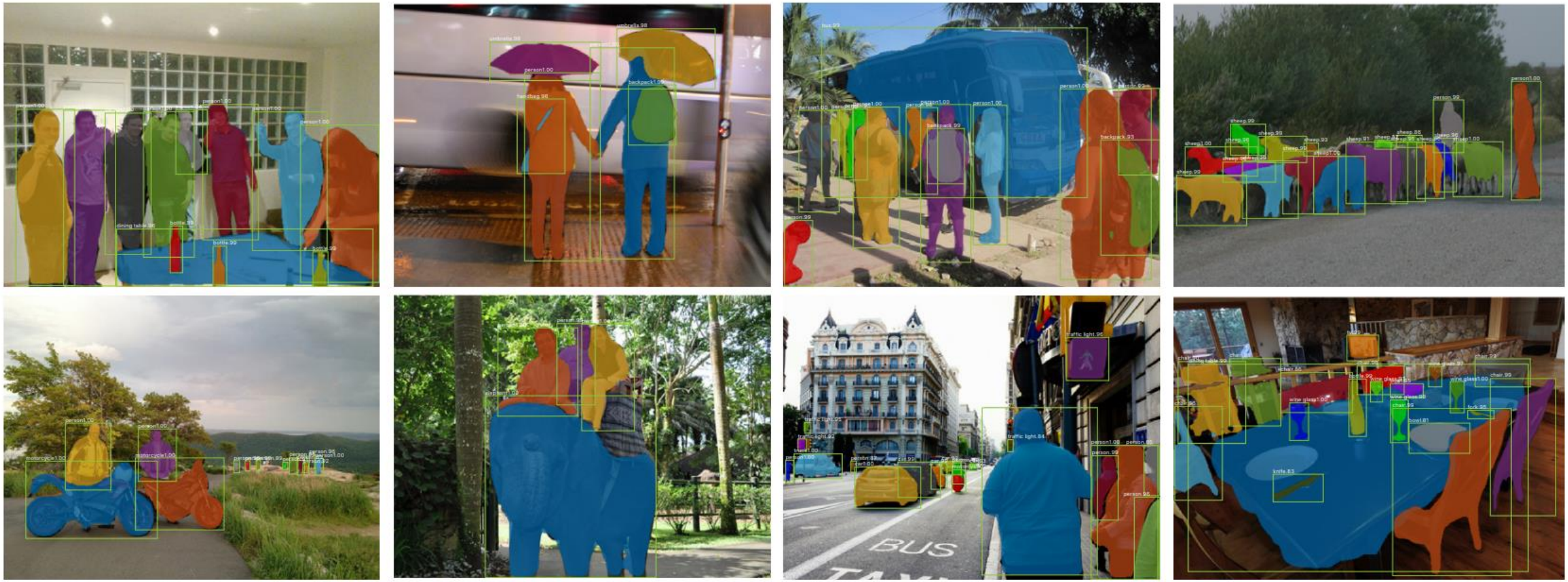
He, Gkioxari, Dollar and Girshick "Mask R-CNN" (ICCV 2017)



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Instance Segmentation: Mask R-CNN

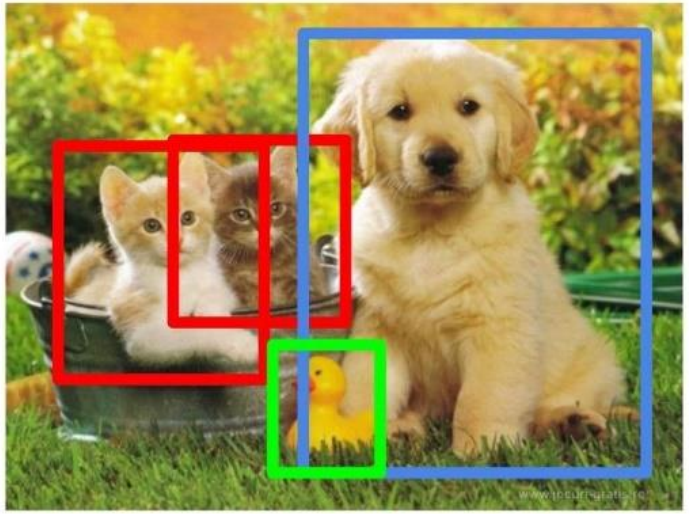
He, Gkioxari, Dollar and Girshick "**Mask R-CNN**" (ICCV 2017)



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Summary

- Deep learning beyond image classification
- Classification “backbones” = “transfer learning”
- Same features - multitasking
- Handling varying number of predictions
- Coping with RF/resolution trade-offs









What's next?

Next week

Monday – Text & Images (Rafail)

Wednesday – Detection and Segmentation II (Lior)